

Meta underwater camera: A smart protocol for underwater image enhancement

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ABSTRACT

Underwater images captured by an underwater camera normally suffer from visual degradation issues, such as color deviations, low contrasts, and detail blurs. Existing studies tend to address these issues separately by individual techniques, such that underwater image visibility can hardly be improved in an overall, consistent manner. To address this limitation, we present a smart protocol for underwater image enhancement. The protocol comprises a comprehensive cascade of seven image enhancement techniques, i.e., attenuated channel compensation, white balance, tone mapping, hue-saturation-lightness (HSL) model-based saturation adjustment, contrast stretching, gamma correction, and high-pass fusion. The protocol is smartly configured by reinforcement learning. Specifically, a set of underwater image multi-color space features are considered as a state, a set of parameter values for the protocol as an action, and an increment of a non-reference visual preference score as a reward. By the reinforcement learning strategy, the parameter values for the seven techniques in the protocol are optimally configured as a whole, resulting in optimal underwater image enhancement results. We refer to the functionality of the smart protocol as a *meta* underwater camera (MUC) methodology because it operates *behind* an underwater camera's operation of capturing underwater images but provides underwater images of high visibility *beyond* those of low visibility originally obtained from the underwater camera. Qualitative and quantitative empirical results validate that our MUC methodology outperforms state-of-the-art underwater image enhancement methods. We release the reproducible code at https://gitee.com/wanghaouc/MUC_UnderwaterImageEnhancement for public evaluations.

1. Introduction

Recently, underwater photogrammetry has become an important research topic for human beings to explore the underwater world. Especially, valuable efforts have been made in underwater camera imaging (Elnashef and Filin, 2019; She et al., 2022), underwater 3D reconstruction (Bruno et al., 2011; Sarafraz and Haus, 2016; Istenić et al., 2020), and underwater photogrammetry modeling (Telem and Filin, 2010, 2013). Among them, underwater camera imaging has been playing a vital role in underwater exploration including ocean energy exploitation, underwater life monitoring, etc (Guo et al., 2020; Wang et al., 2022). High quality underwater images are highly demanded in many practical applications. However, the scattering and absorption effects of the water reduce the visibility of underwater scenes (Yuan et al., 2021). Underwater images captured by underwater cameras normally suffer from color deviations, low contrasts, and detail blurs (Li et al., 2018a). These adverse effects limit the applications of underwater images in practice. To address these limitations, we investigate

an important problem in underwater photogrammetry, i.e., underwater image enhancement. Generally, there are three major families of studies for underwater image enhancement, i.e., physical model based methods, non-physical methods, and data-driven methods. Reviews of them are presented in the subsequent three subsections (i.e., Sections 1.1, 1.2, and 1.3).

1.1. A review of physical model based methods for underwater image enhancement

1.1.1. Motivation

Physical model based methods formulate the formation processes of underwater images subject to physical laws. The latent parameters of an image formation model are estimated with respect to given underwater images by referencing degradation mechanisms. The underwater image enhancement is performed by solving the system of the formation model.

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1.1.2. Literature review

Physical model based methods have been developed for underwater image enhancement. Many studies modified the dark channel prior (DCP) (He et al., 2011) for underwater image enhancement. Chiang and Chen (2012) proposed a wavelength compensation and image dehazing method (WCID). It combines the DCP with the wavelength-dependent compensation method to restore underwater images. Galdran et al. (2015) proposed a red channel method suitable for underwater images. It recovers the lost contrast of an underwater image by recovering the colors associated with short wavelengths. Drews et al. (2016) proposed an underwater dark channel prior (UDCP) method for underwater image enhancement by observing the relevance of the absorption rate in the red channel of underwater images. Peng et al. (2018) proposed a generalized dark channel prior (GDCP) for underwater image enhancement. It uses the depth-dependent color change, scene ambient light differential, and an adaptive color-corrected image formation model to better restore degraded images. Yang et al. (2020b) proposed an underwater image enhancement model based on the dark channel reflection-illumination decomposition and local scattering lighting estimation. It promotes the clarity of edge detail and the colorfulness of the output image. Liang et al. (2022) proposed a Generalized Underwater Dark Channel Prior (GUDCP) for underwater image enhancement. It consists of a backscattered light estimation method based on multiple prior, a transmission estimation method and a discrete distance guided color correction method. In addition, there have been studies employing the simplified Jaffe–McGlamery model (McGlamery, 1980) for underwater image enhancement. Li et al. (2016b) proposed an underwater image enhancement method based on the minimum information loss principle and histogram distribution prior. Wang et al. (2018) proposed an adaptive attenuation-curve prior for underwater image enhancement. It estimates the water light by the smoothness property and the attenuation of light. Zhou et al. (2019) proposed an enhancement method to handle the scattering and absorption problems of light with different wavelengths based on the color-line model and a wavelength-compensation method to correct the color deviations caused by the water depth. Dai et al. (2020) proposed an underwater image enhancement method based on attenuation curve decomposition. It estimates the transmission maps and the background light from a single underwater image by decomposing the attenuation curves on RGB channels and introduces a color balance method to output an image with natural appearance. Berman et al. (2021) proposed an underwater single image color restoration method using haze-lines. It estimates the attenuation ratios of the blue–red and blue–green color channels to reduce the underwater image restoration problems to a single image dehazing. Akkaynak and Treibitz (2018, 2019) revised the simplified Jaffe–McGlamery model and proposed a more complicated underwater image color correction method.

1.1.3. Limitations

Most physical model based methods use simplified underwater formation models. A single physical model based method tends not to be suitable for a variety of underwater scenes, giving rise to underestimated or overestimated specifics in enhancement results (Yuan et al., 2022; Akkaynak et al., 2017; Akkaynak and Treibitz, 2018). This may cause unstable and visually unpleasing results when the individual method is indiscriminately applied to different underwater scenes.

1.2. A review of non-physical methods for underwater image enhancement

1.2.1. Motivation

Non-physical methods consider underwater image enhancement as a pure image processing problem without considering underwater image formation mechanisms. They aim to enhance the color, contrast, and sharpness of underwater images by adjusting pixel values of the degraded images based on existing image processing techniques.

1.2.2. Literature review

Recently, non-physical methods have been developed for improving the quality of underwater images. Many studies applied histogram-based (Garg et al., 2018; Ghani, 2018), retinex-based (Fu et al., 2014; Zhang et al., 2017; Zhuang and Ding, 2020), and fusion-based methods (Ancuti et al., 2012, 2018; Zhang et al., 2022b) to underwater image enhancement. Garg et al. (2018) proposed a strategy for underwater image enhancement using contrast-limited adaptive histogram equalization and percentile methodologies. Ghani (2018) proposed an underwater image contrast enhancement method for increasing the visibility of deep underwater images. It integrates the three main steps of homomorphic filtering, recursive-overlapped contrast limited adaptive histogram specification and dual-image wavelet fusion. Sethi and Sreedevi (2019) proposed a contrast and information enhancement of underwater images (CIEUI) method, which enhances underwater images by improving their contrast and information content using Multi-Objective Particle Swarm Optimization (MOPSO). It effectively corrects color and contrast and enhances information without introducing artifacts. Fu et al. (2014) proposed a retinex-based enhancing approach for a single underwater image. It addresses the fuzz and under-exposure by decomposing the reflectance and illumination from single color corrected underwater image. Zhang et al. (2017) proposed a LAB-MSR method for underwater image enhancement inspired by the retinex framework. It utilizes the combination of the bilateral filter and trilateral filter on the three channels of the image in LAB color space according to the characteristics of each channel. Zhuang and Ding (2020) proposed an edge-preserving filtering retinex algorithm for single underwater image enhancement. It embeds gradient domain guided image filtering priors of reflection and illumination into a retinex-based variational framework for promoting image structures and reducing artifacts. Zhuang et al. (2021) proposed a Bayesian retinex algorithm for enhancing single underwater image. It adopts a color correction to address color deviations and imposes multi-order gradient priors on reflectance and illumination. Zhuang (2021) proposed a variational retinex algorithm for enhancing single underwater image with multi-order gradient priors of reflectance and illumination. It is fast performed on pixel-wise operations without requiring additional prior knowledge about underwater imaging conditions. Zhang et al. (2022a) proposed an underwater image enhancement method based on retinex-inspired color correction and detail preserved fusion technology. It fuses a local contrast-enhanced version, a detail version, and a global contrast-enhanced version of the corrected underwater image to obtain the final enhanced image. Ancuti et al. (2012) proposed an underwater image enhancement method by blending a contrast-enhanced image and a color-corrected image in a multi-scale fusion strategy. Ancuti et al. (2018) modified their work (Ancuti et al., 2012) to enhance underwater images by fusing color-compensated and white-balanced versions of the original degraded image. Zhang et al. (2022b) proposed an underwater image enhancement method via attenuated color channel correction and detail preserved contrast enhancement. It consists of a color correction method, a fusion-based contrast improvement method, and a multi-scale unsharp masking method. Sethi and Sreedevi (2020) proposed a fusion of underwater image enhancement and restoration (FUIER) method, which generates two versions of the single input image and these two versions are fused using Laplacian pyramid-based fusion to get the enhanced image. It efficiently achieves the color deviation removal, contrast enhancement and dehazing. Zhang et al. (2022c) proposed a MILE method for underwater image enhancement. It considers the minimum color loss principle and the maximum attenuation map-guided fusion principle in the color correction, and introduces the integral and squared integral maps to adjust the contrast of the input image. Li et al. (2022b) proposed an underwater image enhancement framework combining variational methods and pyramid technology to improve image quality in the frequency domain. It consists of an adaptive variational contrast enhancement (AVCE) model and a total Laplacian model to enhance the contrast of foreground and preserve texture features at different scales.

1.2.3. Limitations

The parameters of the non-physical methods are normally set to empirical values, which do not always maintain the enhancement performance under varying underwater imaging conditions, and may give rise to under-enhancement or over-enhancement.

1.3. A review of data-driven methods for underwater image enhancement

1.3.1. Motivation

Data-driven models (e.g., deep models) are structured in terms of composites of nonlinear processing units. To guarantee the representational power of the model, large amounts of labeled data are required for training. Specifically, the data-driven underwater image enhancement methods require both underwater images and their ground-truth enhanced images as training data. In this context, a data-driven deep model operates in an end-to-end manner that learns a direct mapping from a degraded image to its corresponding clear scene.

1.3.2. Literature review

Recently, data-driven methods have been developed for underwater image enhancement. Many researchers tried to train the convolutional neural networks (CNNs) to learn the direct mapping from a degraded image to the corresponding clear scene. Li et al. (2020a) proposed an underwater image enhancement convolutional neural network (UWCNN) model based on an underwater scene prior. It can directly reconstruct the clear latent underwater image and be used to synthesize underwater image training data. Li et al. (2020b) proposed an underwater image enhancement model Water-Net. It is a gated fusion network, which fuses the inputs with the predicted confidence maps to achieve the enhanced result. Wu et al. (2021) proposed a two-stage underwater image convolutional neural network (CNN) based on structure decomposition (UWCNN-SD). It includes a preliminary enhancement network for high and low frequency components and a refinement network for clear details. Li et al. (2021) proposed an Ucolor underwater image enhancement network via medium transmission-guided multi-color space embedding. It uses an attention mechanism to extract the most discriminative features from multiple color spaces. Jiang et al. (2021) proposed a Lightweight Cascaded Network (LCNet) based on Laplacian image pyramids for underwater image enhancement. It implements cascaded blocks upon a residual network at each pyramid level. Huang et al. (2022) proposed an adaptive group attention (AGA), which can dynamically select visually complementary channels based on the dependencies, reducing the number of further attention parameters. The AGA is applied in the Swin Transformer module and used to design an end-to-end underwater image enhancement network. Jiang et al. (2022) proposed a target oriented perceptual adversarial fusion network (TOPAL). It consists of two manually-crafted structures (i.e., Multi-scale Dense Boosted (MDB) and Deep Aesthetic Render (DAR)) to perform the muddy restoration and color correction, respectively. Qi et al. (2022) proposed an underwater image co-enhancement network (UICoE-Net) based on an encoder-decoder Siamese architecture. It introduces correlation feature matching units into the multiple layers of the Siamese encoder-decoder structure to communicate the mutual correlation of the two branches. In addition, many researchers tried to use generative adversarial networks (GANs) for underwater image enhancement tasks. Chen et al. (2019) proposed a GAN-based restoration scheme (GAN-RS) for adaptively restoring underwater images in real time. It restores underwater visual quality using a single-shot network for higher computational efficiency and greater restoration quality. Yang et al. (2020a) proposed a conditional generative adversarial network (cGAN) for underwater image enhancement. It uses a multi-scale generator for extracting larger receptive fields, and a dual discriminator pushed the generator towards better visual quality. Liu et al. (2020) proposed a deep multi-scale feature fusion net based on the conditional generative adversarial network (MLFcGAN) for underwater image color correction. It extracts multi-scale features

and augmented local features on each scale with global features. Is-lam et al. (2020a) proposed a fast underwater image enhancement conditional GAN-based model (FUnIE-GAN). It formulates a perceptual loss function by evaluating image quality based on its global color, content, local texture, and style information. Ye et al. (2020) proposed an unsupervised adaptation network to jointly estimate scene depth and correct the color from a single underwater image. It considers the problem as those of style-level and feature-level adaptation. Zhou et al. (2022) proposed a domain adaptive learning framework based on physical model feedback for underwater image enhancement. It uses the image degradation model as a physical constrain feedback of framework, and the model provides explicit constraints for the problem of underwater image enhancement. Lin et al. (2022) proposed a conditional generative adversarial network with dual-branch progressive generator. The generator consists of two independent branches and a progressive enhancement algorithm.

1.3.3. Limitations

Most data-driven underwater image enhancement methods are trained with synthetic datasets (Li et al., 2018b; Ueda et al., 2019; Dudhane et al., 2020), due to the lack of real underwater images and their corresponding ground-truth non-water images. In this scenario, the effectiveness of the data-driven methods is limited by the quality of synthetic images, which may neither comprehensively characterize various underwater imaging scenarios (Berman et al., 2021) nor satisfy human perception.

1.4. Our novel underwater image enhancement framework

From the reviews presented in Sections 1.1, 1.2, and 1.3, we observe that there are three unresolved challenges to underwater image enhancement as follows:

(a) In order to avoid indiscriminately using one single technique for enhancing underwater images obtained under different imaging conditions, a comprehensive selection of image enhancement techniques is required. They should be complementary to one another and operate as a whole to address the varying underwater degrading scenarios.

(b) In order to avoid empirically setting the parameter values for the underwater image enhancement techniques, a smart configuration method is required. It should provide optimal parameter values for the underwater image enhancement techniques.

(c) In order to avoid misleadingly enhancing underwater images into unsatisfying results (e.g., those limited by synthetic images), an underwater image enhancement implementation scheme that follows human visual perception is required. It should encourage the underwater image enhancement to operate in favor of human observation of underwater scenes.

To address these challenges, we make three novel contributions as follows:

(a) **Comprehensive protocol:** We develop a comprehensive protocol for underwater image enhancement. The protocol comprises a comprehensive cascade of seven image enhancement techniques including attenuated channel compensation, white balance, tone mapping, HSL model-based saturation adjustment, contrast stretching, gamma correction, and high-pass fusion. The protocol comprehensively solves the problems of quality degradation in underwater images. The first four techniques address the problem of color deviations, the fifth and sixth techniques address the problem of low contrasts, and the seventh technique addresses the problem of detail blurs. Therefore, the seven techniques are complementary to one another such that the performance that one technique cannot achieve can be achieved by the other techniques. The protocol comprising the seven techniques operates as a whole and achieves comprehensive underwater image enhancement under varying imaging conditions.

(b) **Smart configuration:** We develop a reinforcement learning strategy for configuring the parameter values for the comprehensive

protocol for underwater image enhancement. The parameter configuration does not rely on empirical setting but is performed smartly in terms of the reinforcement learning strategy that solely relies on the investigated underwater images. The reinforcement learning strategy upgrades the protocol into a smart protocol which achieves its best performance of enhancing underwater images.

(c) **Visually preferable implementation:** We develop an implementation scheme for the smart protocol with respect to a visually preferable score. We consider the increments of a non-reference visual preference score (which is a combination of two commonly used non-reference underwater image quality metrics i.e., underwater color image quality evaluation (UICQE) (Yang and Sowmya, 2015) and underwater image quality measure (UIQM) (Panetta et al., 2016)) as rewards for the reinforcement learning. The UICQE and UIQM are inspired by the mechanisms of human visual systems, and they encourage the smart protocol to enhance images in favor of human perception.

We refer to the functionality of the smart protocol as a meta underwater camera (MUC) methodology because it operates behind an underwater camera's operation of capturing underwater images but provides underwater images of high visibility beyond those of low visibility originally obtained from the underwater camera. Comprehensive qualitative and quantitative empirical results validate that our MUC methodology outperforms state-of-the-art underwater image enhancement methods.

1.5. Paper organization

The remainder of this paper is organized as follows. Section 2 presents the comprehensive protocol for underwater image enhancement. Section 3 presents the smart configuration of the comprehensive protocol with reinforcement learning. Section 4 presents the methodology of meta underwater camera. Section 5 presents the experiments and analyses the results. Section 6 concludes the paper.

2. A comprehensive protocol for underwater image enhancement

This section presents a comprehensive protocol for underwater image enhancement to solve the three problems of color deviations, low contrasts, and detail blurs with underwater images. The protocol comprises a comprehensive cascade of seven image enhancement techniques including attenuated channel compensation, white balance, tone mapping, HSL model-based saturation adjustment, contrast stretching, gamma correction, and high-pass fusion, as shown in Fig. 1. Different from Sethi and Sreedevi (2020), Li et al. (2022a), Hou et al. (2020), our protocol uses attenuated channel compensation, white balance, tone mapping, and HSL model-based saturation adjustment (i.e., the first four techniques of the protocol) to solve the problem of color deviations, uses contrast stretching and gamma correction (i.e., the fifth and sixth techniques of the protocol) to solve the problem of low contrasts, and uses high-pass fusion (the seventh technique of the protocol) to solve the problem of detail blurs, establishing a protocol that operates as a whole and achieves comprehensive underwater image enhancement under varying imaging conditions. The seven image enhancement techniques and the overall protocol will be presented in the subsequent subsections (i.e., Sections 2.1 and 2.2).

2.1. Seven techniques for solving the problems of color deviations, low contrasts, and detail blurs

(a) **Attenuated channel compensation:** The technique of attenuated channel compensation processes an original underwater image I_c to a compensated image I_c^a . The original three-channel underwater image I_c consists of three single channel images, i.e., the red, green and blue channel images I_r , I_g and I_b . The range of intensity values for each channel image is $[0, 1]$. Let $\bar{I}_r$, $\bar{I}_g$, and $\bar{I}_b$ denote the means for the intensity values in the red, green, and blue channels over

the whole image plane, respectively. For a turbid-water scene, the green channel is relatively well preserved in images, compared to the red and blue channel images (Song et al., 2021). Red light with the longest wavelength is significantly attenuated in underwater scenes, and the intensity values in the red channel image are small. The blue channel image may be considerably attenuated in the places with a high concentration of plankton because of the absorption by organic matter (Lu et al., 2015). In this case, $\bar{I}_g$ is greater than $\bar{I}_b$, i.e., $\bar{I}_g > \bar{I}_b$. Motivated by the observations in the previous underwater study (Ancuti et al., 2018), we use the green channel image I_g to compensate for the attenuation of the red and blue channel images I_r and I_b . In addition, for a deep-water scene, the blue channel is relatively well preserved in images, compared to the red and green channel images (Zhang et al., 2022b). When light propagates in water, the absorption of light by the water depends on the wavelength of light. Red light disappears first, then green light, and after that blue light (Akkaynak et al., 2017). In this case, $\bar{I}_b$ is greater than or equal to $\bar{I}_g$, i.e., $\bar{I}_b \geq \bar{I}_g$. We use the blue channel image I_b to compensate for the attenuation of the green channel image I_g . Different from the turbid-water scene, we use the compensated green channel image I_g^a to compensate for the attenuation of the red channel image I_r because the green channel image is the one that contains opponent color information compared to the red channel image, and using the green channel image allows better recovering the entire color spectrum while maintaining the natural appearance of the background (Ancuti et al., 2018). Generalizing the above observations, we formulate the technique of attenuated channel compensation as follows:

$$\begin{cases} I_b^a(x, y) = I_b(x, y) + \alpha(1 - I_b(x, y))(\bar{I}_g - \bar{I}_b)I_g(x, y)\mathbb{1}[\bar{I}_g > \bar{I}_b], \\ I_g^a(x, y) = I_g(x, y) + \alpha(1 - I_g(x, y))(\bar{I}_b - \bar{I}_g)I_b(x, y)\mathbb{1}[\bar{I}_b \geq \bar{I}_g], \\ I_r^a(x, y) = I_r(x, y) + \beta(1 - I_r(x, y))(\bar{I}_g^a - \bar{I}_r)I_g^a(x, y), \end{cases} \quad (1)$$

where α and β are parameters that control the compensation intensity. Big values of α and β imply high compensation intensity. $\mathbb{1}[\cdot]$ is a condition function. When $[\cdot]$ is true, $\mathbb{1}[\cdot]$ is equal to 1, and when $[\cdot]$ is false, $\mathbb{1}[\cdot]$ is equal to 0. I_b^a , I_g^a , and I_r^a are compensated blue, green, and red channel images, respectively. $\bar{I}_g^a$ is the mean value for the compensated green channel image over the whole image plane. x and y are the position coordinates of each pixel.

Based on the technique of attenuated channel compensation, the compensated image I_c^a is obtained.

(b) **White balance:** The technique of white balance processes the compensated image I_c^a to a balanced image I_c^b . We use the Gray-World assumption (Buchsbbaum, 1980) to estimate and balance color deviations for the compensated image I_c^a . The balanced red, green and blue channel images I_r^b , I_g^b and I_b^b are expressed as follows:

$$\begin{aligned} I_r^b(x, y) &= \frac{\bar{I}_c^a}{\bar{I}_r^a} I_r^a(x, y), \\ I_g^b(x, y) &= \frac{\bar{I}_c^a}{\bar{I}_g^a} I_g^a(x, y), \\ I_b^b(x, y) &= \frac{\bar{I}_c^a}{\bar{I}_b^a} I_b^a(x, y), \end{aligned} \quad (2)$$

where $\bar{I}_c^a$ is the mean value for all the intensity values in the three channels of the whole compensated image I_c^a . $\bar{I}_r^a$, $\bar{I}_g^a$, and $\bar{I}_b^a$ are the mean values for the compensated red, green, and blue channels I_r^a , I_g^a , and I_b^a over the whole image plane, respectively.

Based on the technique of white balance, the balanced image I_c^b is obtained.

(c) **Tone mapping:** The technique of tone mapping processes the balanced image I_c^b to a tone-mapped image I_c^c . The intensity values for the balanced image I_c^b may be greater than one. There are two existing processing methods. One is to set all the intensity values greater than one to one, and this method may give rise to imbalanced image color. The other is to obtain the minimum and maximum intensity values

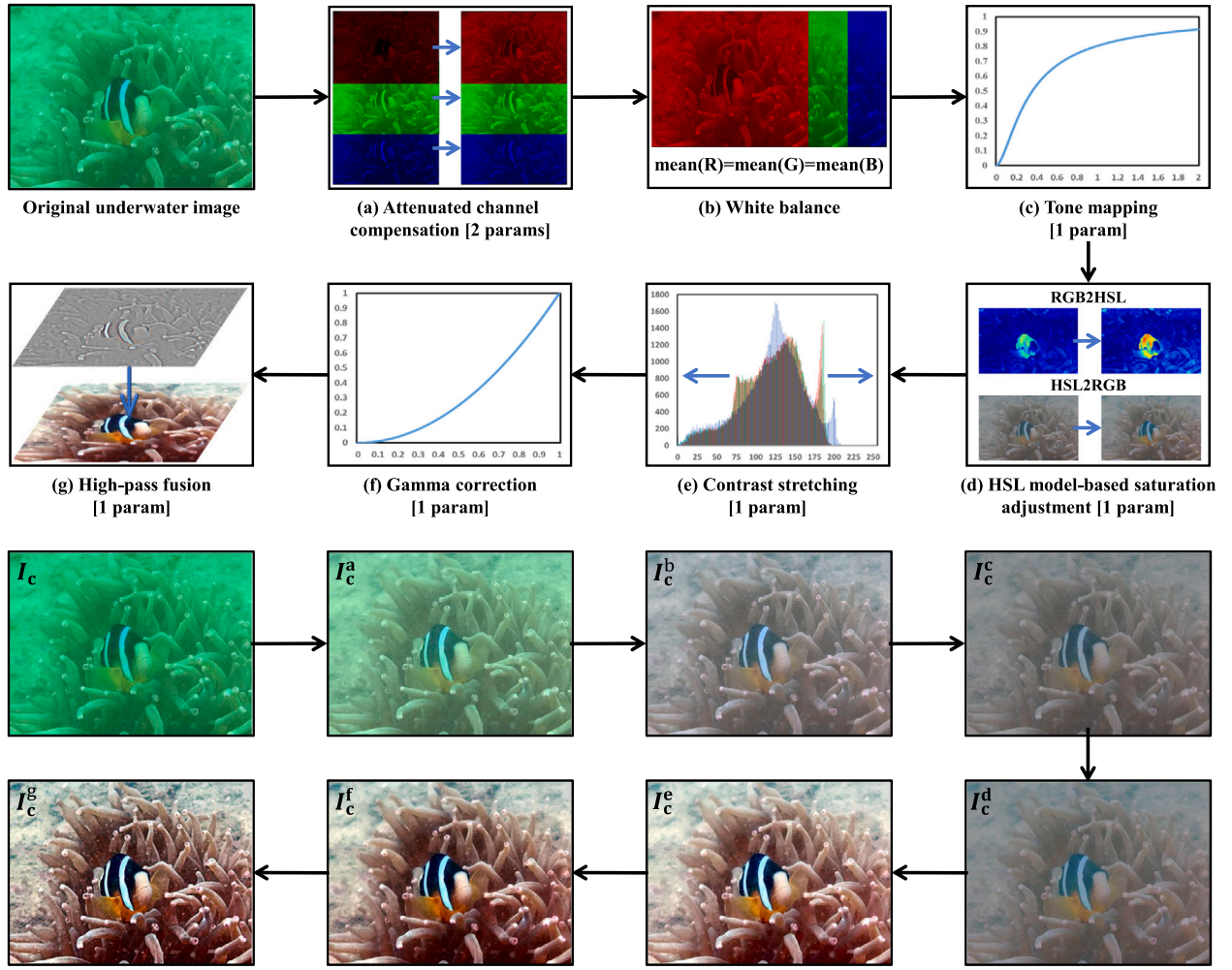


Fig. 1. A comprehensive protocol for underwater image enhancement. [# param(s)] indicates the number of parameters in one technique.

and linearly scale all intensity values to [0,1], and this method may give rise to a dark image. To address these limitations, we use the ACES filmic tone mapping curve (Krzysztof, 2016) to tone mapping. The tone-mapped image I_c^c are expressed as follows:

$$I_c^c(x, y) = \frac{\delta I_c^b(x, y)(\delta \eta_0 I_c^b(x, y) + \eta_1)}{\delta I_c^b(x, y)(\delta \eta_2 I_c^b(x, y) + \eta_3) + \eta_4}, \quad (3)$$

where δ is a parameter that controls the brightness of the tone-mapped image I_c^c . A big value of δ implies high brightness. $\eta_0, \eta_1, \eta_2, \eta_3$, and η_4 are the balance parameters.

Based on the technique of tone mapping, the tone-mapped image I_c^c is obtained.

(d) **HSL model-based saturation adjustment:** The technique of HSL model-based saturation adjustment processes the tone-mapped image I_c^c to an adjusted image I_c^d . The tone-mapped image I_c^c is close to solving the problem of color deviations but has the limitation of low color saturation. To address this limitation, the tone-mapped image I_c^c is converted from the RGB color model to the HSL color model. The HSL color model comprises hue, lightness, and saturation channel images. The saturation channel image I_s^c of the tone-mapped image is adjusted as follows:

$$I_s^d(x, y) = (1 + \xi) I_s^c(x, y), \quad (4)$$

where ξ is a parameter that controls the color saturation adjustment intensity. A big value of ξ implies high color saturation adjustment intensity. I_s^d is the adjusted saturation channel image.

After the saturation channel image is adjusted, the adjusted image I_c^d is obtained by converting from the HSL color model to the RGB color model.

Based on the technique of HSL model-based saturation adjustment, the adjusted image I_c^d is obtained.

(e) **Contrast stretching:** The technique of contrast stretching processes the adjusted image I_c^d to a stretched image I_c^e . The range of intensity values for the adjusted image I_c^d is (0,1), and the adjusted image I_c^d has low contrasts. To address this limitation, we use contrast stretching (Gillespie, 1992) to extend the gray dynamic range and improve the image contrast. The intensity values for the adjusted image I_c^d are clipped to control the strength of the contrast stretching, and the clipped red, green, and blue channel images I_r^d, I_g^d , and I_b^d of the adjusted image I_c^d are expressed as follows:

$$\begin{aligned} I_r^d(x, y) &= \text{clip}(I_r^d(x, y), I_r^d(\lambda\%), I_r^d(1 - \lambda\%)), \\ I_g^d(x, y) &= \text{clip}(I_g^d(x, y), I_g^d(\lambda\%), I_g^d(1 - \lambda\%)), \\ I_b^d(x, y) &= \text{clip}(I_b^d(x, y), I_b^d(\lambda\%), I_b^d(1 - \lambda\%)), \end{aligned} \quad (5)$$

where λ is a parameter that controls the range of the image contrast stretching. A big value of λ implies high contrast. $I_r^c(\lambda\%), I_g^c(\lambda\%)$ and $I_b^c(\lambda\%)$ are the largest intensity values for the top $\lambda\%$ of the I_r^c, I_g^c and I_b^c . $I_r^c(1 - \lambda\%), I_g^c(1 - \lambda\%)$ and $I_b^c(1 - \lambda\%)$ are the largest intensity values for the top $(1 - \lambda\%)$ of the I_r^c, I_g^c and I_b^c . $\text{clip}(I, I_{\min}, I_{\max})$ is a function that clips the array I to the range $[I_{\min}, I_{\max}]$, i.e., values less than $I_{\min}$ become $I_{\min}$, and values greater than $I_{\max}$ become $I_{\max}$.

We use a simple yet effective linear contrast stretching technique, and the stretched red, green, and blue channel images I_r^e , I_g^e , and I_b^e are expressed as follows:

$$\begin{aligned} I_r^e(x, y) &= \frac{I_r^{d'}(x, y) - I_r^d(\lambda\%)}{I_r^d(1 - \lambda\%) - I_r^d(\lambda\%)}, \\ I_g^e(x, y) &= \frac{I_g^{d'}(x, y) - I_g^d(\lambda\%)}{I_g^d(1 - \lambda\%) - I_g^d(\lambda\%)}, \\ I_b^e(x, y) &= \frac{I_b^{d'}(x, y) - I_b^d(\lambda\%)}{I_b^d(1 - \lambda\%) - I_b^d(\lambda\%)}. \end{aligned} \quad (6)$$

Based on the technique of contrast stretching, the stretched image I_c^e is obtained.

(f) **Gamma correction:** The technique of gamma correction processes the stretched image I_c^e to a corrected image I_c^f . We use a non-linear curve (i.e., gamma curve) to further correct the brightness and contrast of the stretched image I_c^e . The corrected image I_c^f is expressed as follows:

$$I_c^f(x, y) = I_c^e(x, y)^\gamma, \quad (7)$$

where γ is a parameter that controls the brightness and contrast. For γ greater than one, the contrast of the image in the low grayscale area is reduced, the contrast of the image in the high grayscale area is increased, and the image becomes darker. For γ less than one, the contrast of the image in the high grayscale area is reduced, the contrast of the image in the low grayscale area is increased, and the image becomes brighter.

Based on the technique of gamma correction, the corrected image I_c^f is obtained.

(g) **High-pass fusion:** The technique of high-pass fusion processes the corrected image I_c^f to a fused image I_c^g . The corrected image I_c^f suffers from detail blurs. To address this limitation, we use high-pass fusion to improve the sharpness of the corrected image I_c^f and enrich the details. Firstly, the corrected image I_c^f is smoothed with a Gaussian filter to obtain a smoothed image. Secondly, the corrected image I_c^f subtracts the smoothed image to obtain an edge-enhanced high-pass mask I_m as follows:

$$I_m = I_c^f - F_{\text{Gauss}}(I_c^f, \tau) + \phi, \quad (8)$$

where τ is the standard deviation for the Gaussian kernel. A big value of τ implies high smoothness. $F_{\text{Gauss}}(\cdot)$ is the Gaussian filter, $F_{\text{Gauss}}(I_c^f, \tau)$ is the smoothed image, ϕ is a balance parameter that to keep the intensity values for the high-pass mask I_m .

Finally, the corrected image I_c^f is fused with the high-pass mask I_m to obtain the fused image I_c^g as follows:

$$I_c^g(x, y) = \begin{cases} \frac{I_c^f(x, y)I_m(x, y)}{\phi}, & I_m(x, y) < \phi; \\ 1 - \frac{(1 - I_c^f(x, y))(1 - I_m(x, y))}{\phi}, & \text{otherwise.} \end{cases} \quad (9)$$

Based on the technique of high-pass fusion, the fused image I_c^g is obtained.

2.2. The overall comprehensive protocol for underwater image enhancement

The seven techniques (i.e., attenuated channel compensation, white balance, tone mapping, HSL model-based saturation adjustment, contrast stretching, gamma correction, and high-pass fusion) presented in Section 2.1 form a comprehensive protocol for underwater image enhancement to solve the three problems of color deviations, low contrasts, and detail blurs. Among them, the first four techniques are to solve the problem of color deviations. According to the Gray-world assumption (Buchsbbaum, 1980), all color channels of an image without color deviations have similar mean values. Existing studies (Sethi and

Sreedevi, 2017; Ancuti et al., 2018; Song et al., 2021) have validated that the white balance technique (i.e., Gray-world assumption (Buchsbbaum, 1980)) is effective in neutralizing color deviations in underwater images. However, using the white balance technique directly on underwater images normally suffers from artifacts (e.g., red artifacts). Those red artifacts arise from a very small mean value for the red channel image, leading to an overcompensation of this channel image in locations where the red component is present (mainly because the Gray-world hypothesis proposes to divide the intensity values of each channel image by its mean value). Different from Sethi and Sreedevi (2019), Zhang et al. (2021), Yan et al. (2022), our protocol uses the attenuated channel compensation technique that has the potential of correcting color deviations without introducing artifacts. Therefore, the first technique (i.e., attenuated channel compensation) is undertaken before the second technique (i.e., white balance). The white balance technique makes all channels of the image have similar mean values, and the intensity values for the balanced image may be greater than one. There are two existing methods for constraining the intensity values to be smaller than one. One is to set all the intensity values greater than one to one, and this method may give rise to imbalanced image color. The other is to linearly scale all intensity values to [0,1] with respect to the original minimum and maximum intensity values, and this method may give rise to a dark image. To maintain color balance, the third technique (i.e., tone mapping of ACES filmic tone mapping curve) is used. It non-linearly scales all intensity values to (0,1) and maintains color balance. Furthermore, underwater images tend to have low saturation, which is not in line with human visual perception (Yang and Sowmya, 2015; Panetta et al., 2016). The fourth technique (i.e., HSL model-based saturation adjustment) is a simple, fast and effective technique to improve saturation. The fifth and sixth techniques are to solve the problem of low contrasts. The third technique (i.e., tone mapping) maps the intensity values of an image to (0,1), where the image may have low gray dynamic range and low contrasts. To stretch the gray dynamic range of the image within [0,1] and improve the contrasts, the fifth technique (i.e., contrast stretching), which is simple yet effective, is used. Furthermore, existing studies (Li et al., 2020b; Sun et al., 2022) have validated that gamma curve as a nonlinear curve is effective to correct the brightness and contrasts of underwater images. Therefore, the sixth technique (i.e., gamma correction) is used after the linear contrast stretching technique, for the purpose of further refining image contrasts. The seventh technique is to solve the problem of detail blurs. Here, the seventh technique (i.e., high-pass fusion) plays an effective role in maintaining the sharpness of the image. It also exhibit better efficiency than other method for the similar purposes (Ancuti et al., 2018; Zhang et al., 2022b).

The seven techniques are complementary to one another for the underwater image enhancement task such that the comprehensive protocol solves the problems of color deviations, low contrasts, and detail blurs in a consistent, overall manner. We will further validate the order of the techniques for maintaining the underwater image enhancement performance in Section 5.10.1.

In addition, in the protocol there are parameters, e.g., α and β in Eq. (1), δ in Eq. (3), ξ in Eq. (4), λ in Eq. (5), γ in Eq. (7), and τ in Eq. (8), that require reasonably tuned. These parameters directly affect the enhancement performance of the protocol but are normally empirically set to default values which do not guarantee the optimal underwater image enhancement. To address this limitation, the comprehensive protocol is smartly configured by reinforcement learning. This will be presented in the subsequent section (i.e., Section 3).

3. Smart configuration of the comprehensive protocol with reinforcement learning

This section describes the process of smartly configuring the comprehensive protocol via reinforcement learning to construct a smart

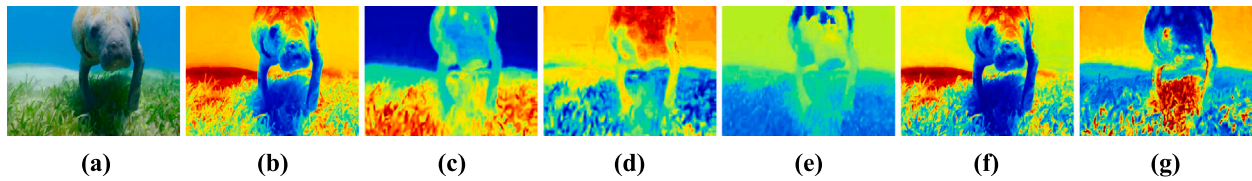


Fig. 2. A visual example of the channels extracted from different color spaces. (a) RGB image, (b) Hue channel image, (c) Saturation channel image, (d) Lightness channel image, (e) Luminance channel image, (f) A channel image, and (g) B channel image. For visualization, the channels in (b)–(g) are represented by heatmaps, where the color ranging from the blue to red represents the value from small to large. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Key component specifications for the reinforcement learning.

Component	Specification
State	Underwater image multi-color space features
Action	A set of parameter values for the protocol
Reward	Non-reference visual preference score increment

protocol. The smart protocol can achieve the optimal parameter value configuration for underwater image enhancement.

The state, action and reward are three key components for reinforcement learning. To configure the parameter values for the protocol via reinforcement learning, we specify the three components according to the scheme presented in Table 1.

The subsequent subsections (i.e., Sections 3.1–3.3) will present the details of these three components.

3.1. Underwater image multi-color space features as a state

We consider the multi-color space features of one underwater image as a state s . The features of one image in different color spaces represents the image from different aspects. In our work, we exploit the image features in three color spaces, i.e., RGB, HSL, and LAB, as shown in Fig. 2. Specifically, the RGB features straightforwardly reflect the image color characteristics. However, the three channels R, G, and B have high correlations and are easy to be affected by the changes of lightness, shadow, and other factors. By contrast, the HSL features can intuitively reflect the hue, saturation, and lightness of the image. Additionally, the LAB features render better distributed color representations for human perception. In this scenario, the color difference between two pixels may be small in one color space and large in another. It is observed (Li et al., 2021) that the three color spaces involve almost all aspects related to image degradation, i.e., color, hue, saturation, intensity, and luminance. Therefore, the multi-color space features of RGB, HSL, and LAB can effectively facilitate the color deviations in underwater images. We will analyze the contribution of each color space in Section 5.10.4. We use three resnet18 networks without the final fully connected layers as the feature extraction to extract three 512-dimensional features of the RGB color space, the LAB color space, and the HSL color space, separately. The three 512-dimensional features are concatenated as the state s .

3.2. A set of parameter values for the protocol as an action

We consider a set of parameter values for the smart protocol as an action a . The set of parameters of the smart protocol includes α , β , δ , ξ , λ , γ , and τ , which form an action $a = \{\alpha, \beta, \delta, \xi, \lambda, \gamma, \tau\}$. The default values for these parameters are set as a baseline action $a_0 = \{\alpha_0, \beta_0, \delta_0, \xi_0, \lambda_0, \gamma_0, \tau_0\}$.

3.3. Non-reference visual preference score increment as a reward

We consider a non-reference visual preference score increment as a reward r . The non-reference visual preference score is a combination of two commonly used non-reference underwater image quality metrics, i.e., UCIQE (Yang and Sowmya, 2015) and UIQM (Panetta et al., 2016). UCIQE is a linear combination of chroma, saturation, and contrast based on CIELab color space. It quantifies the color deviations, low contrasts and blurring that characterize underwater images. A big value of the UCIQE score SC_{UCIQE} implies that the result has a high balance among the chroma, saturations, and contrasts. UIQM comprises a colorfulness measure (UICM), a contrast measure (UIConM), and a sharpness measure (UISM). It follows human visual systems (HVSs). A big value of the UIQM score SC_{UIQM} implies that the result has good color, contrasts, and details and a high consistent with human visual perception. These two metrics quantitatively characterize the color deviations, low contrasts, and detail blurs in underwater images in different spaces. We combine the two metrics to comprehensively characterize underwater image quality. The combination comprehensively addresses the problems of the color deviations, low contrasts, and detail blurs such that it improves the visual performance over the commonly adopted image quality evaluation metrics. The non-reference visual preference score SC_{VP} of the image I_c is given as follows:

$$SC_{VP}(I_c) = SC_{UCIQE}(I_c) + \sigma SC_{UIQM}(I_c), \quad (10)$$

where σ is a balance parameter.

In addition, we use the default values for parameters as the baseline action $a_0 = \{\alpha_0, \beta_0, \delta_0, \xi_0, \lambda_0, \gamma_0, \tau_0\}$ to configure the protocol and obtain a baseline image E_c^0 . The visual preference score of the baseline image E_c^0 is computed according to Eq. (10) as a baseline score $SC_{VP}(E_c^0)$. The reward r is defined as the increment from the baseline score $SC_{VP}(E_c^0)$ to the enhanced score $SC_{VP}(E_c)$ as follows:

$$r = SC_{VP}(E_c) - SC_{VP}(E_c^0). \quad (11)$$

3.4. Reinforcement learning of the smart protocol

The comprehensive protocol is smartly configured via reinforcement learning to construct a smart protocol, as shown in Fig. 3. This subsection describes how to train a reinforcement learning network, which takes the original underwater image features as input and gives optimal parameter values. Given the original underwater image I_c , the feature extraction extracts the features of the original underwater image I_c as the state s . The reinforcement learning network takes the state s as input and outputs the action $a = \{\alpha, \beta, \delta, \xi, \lambda, \gamma, \tau\}$. The protocol takes the original underwater image I_c and the action $a = \{\alpha, \beta, \delta, \xi, \lambda, \gamma, \tau\}$ as input and gives the enhanced image E_c as output. The visual preference score of the enhanced image E_c is computed according to Eq. (10) as the enhanced score $SC_{VP}(E_c)$. The protocol takes the original underwater image I_c and the baseline action $a_0 = \{\alpha_0, \beta_0, \delta_0, \xi_0, \lambda_0, \gamma_0, \tau_0\}$ as input and gives a baseline image E_c^0 as output. The visual preference score of the baseline image E_c^0 is computed according to Eq. (10) as the baseline score $SC_{VP}(E_c^0)$. The reward r is computed according to

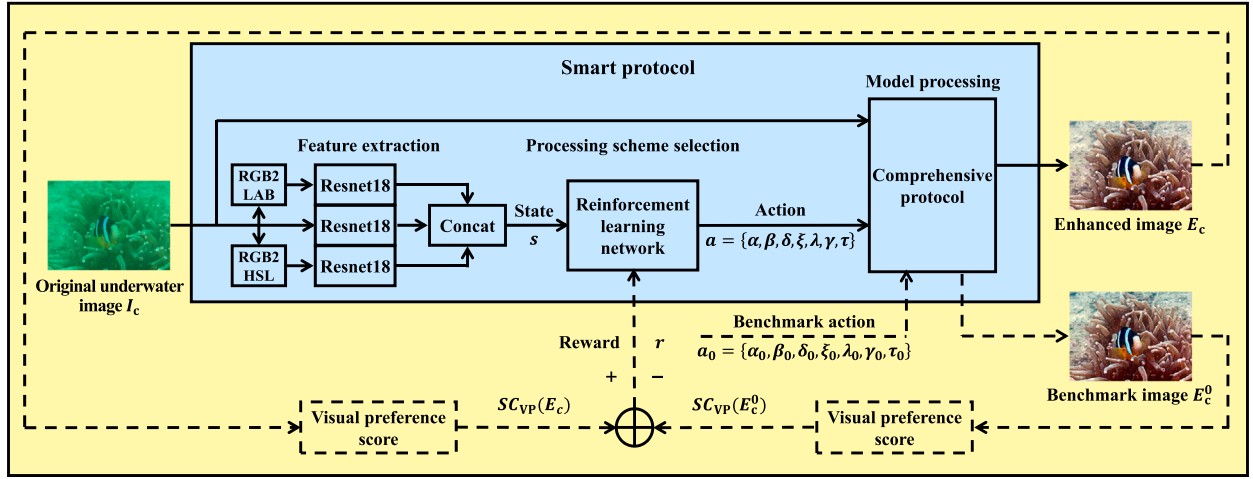


Fig. 3. Reinforcement learning of the smart protocol.

Eq. (11). We follow the proximal policy optimization (PPO) (Schulman et al., 2017) to conduct reinforcement learning with respect to Eq. (11). Given a previously unseen underwater image, the learned reinforcement learning network can provide optimal parameter values, which are those the most suitable for the task of enhancing the given underwater image, for the comprehensive protocol.

We refer to the comprehensive protocol that is smartly configured by reinforcement learning as the smart protocol, and use it for underwater image enhancement.

4. Meta underwater camera

We use the smart protocol learned according to Section 3.4 to upgrade an underwater camera to a meta underwater camera (MUC), as shown in Fig. 4. The smart protocol operates behind an underwater camera's operation of capturing underwater images but provides underwater images of high visibility beyond those of low visibility originally obtained from the underwater camera. Specially, the smart protocol takes the original underwater image captured by the underwater camera as input and gives the enhanced image as output. Therefore, the meta underwater camera is an end to end methodology for capturing clear underwater scenes.

5. Empirical evaluations

This section empirically validates the effectiveness of our meta underwater camera methodology. Specifically, the two datasets for validation are presented in Section 5.1, the evaluation metrics are presented in Section 5.2, the experiment setup is presented in Section 5.3, the methods for comparison are presented in Section 5.4, the comprehensively empirical evaluations on the two datasets are presented in Sections 5.5 and 5.6, the evaluations of the runtime are presented in Section 5.8, the potential applications of the meta underwater camera methodology are presented in Section 5.9, and the ablation study of the meta underwater camera methodology is presented in Section 5.10.

5.1. Datasets

We conduct experiments on two underwater image datasets (i.e., UIEB (Li et al., 2020b) and U45 (Islam et al., 2020b)). The UIEB dataset consists of 950 underwater images (890 real-scenes underwater images and 60 challenge underwater images) that cover different quality features and contents in various underwater scenes. The U45 consists of 45 real-scenes underwater images, including three subsets (i.e., green, blue, and haze-like). They aim to implement a comprehensive perceptual study and analysis for underwater image enhancement methods.

5.2. Evaluation metrics

We evaluate the underwater image quality with five metrics, i.e., UCIQE (Yang and Sowmya, 2015), CCF (Wang et al., 2018a), UIQM (Panetta et al., 2016), PCQI (Yuan et al., 2021), and BCRA (Yuan et al., 2022). The UCIQE evaluates underwater image quality through color density, saturation, and contrast. The CCF combines the colorfulness metric, contrast metric, and haze density metric to predict the color loss caused by absorption, the blurring caused by forwarding scattering, and the haze caused by backward scattering, respectively. The UIQM is an evaluation metric composed of underwater image color (UICM), underwater image sharpness measure (UISM), and underwater image contrast measure (UIConM). The PCQI is an image contrast metric used to evaluate image contrast distortion (before and after enhancement) in the human visual system. The BCRA evaluates the increase of the image detailed information before and after enhancement. It has two scores, i.e., BC(e), which indicates the increase in the number of visible edges, and BC(r), which indicates the enhancement of gradient values for edge pixels. When an underwater image enhancement method has a good performance, the five metrics present high evaluation values.

5.3. Experimental setup

Our MUC methodology is implemented on a workstation with Intel (R) Xeon (R) Gold 5218R CPU @ 2.1G of 32 cores, 376G RAM, and an NVIDIA GeForce RTX 2080Ti GPU. The operating system is CentOS 7. We use the following parameters to implement the reinforcement learning of the smart protocol. Specifically, the learning rate is 3×10^{-4} , the batch size is 8, the discount factor for rewards is 0.9, the entropy coefficient is 0.01, the parameter for the clipped surrogate objective is 0.2, and the Adam optimizer is used. The balance parameters $\eta_0, \eta_1, \eta_2, \eta_3$, and η_4 in Eq. (3) are 2.51, 0.03, 2.43, 0.59, and 0.14, respectively. The balance parameter ϕ in Eq. (9) is 0.5. The balance parameter σ in Eq. (10) is 0.65. The ranges of parameters $\alpha, \beta, \delta, \xi, \lambda, \gamma$, and τ are [1, 1.5], [1, 1.5], [0.6, 1], [0.1, 0.5], [1, 2], [0.7, 1.3], and [3, 10], respectively. The default values for the parameters $\alpha_0, \beta_0, \delta_0, \xi_0, \lambda_0, \gamma_0$, and τ_0 are 1, 1, 0.6, 0.1, 1, 1.3, and 3, respectively. We resize each image into 256×256 patches in both training and testing. The reinforcement learning network, i.e., the agent, consists of three fully-connected layers. The input sizes are 1536, 512, and 128, separately. The output sizes are 512, 128, and 7, separately. Each layer except the final one is followed by a layer normalization and a leaky relu activation function.

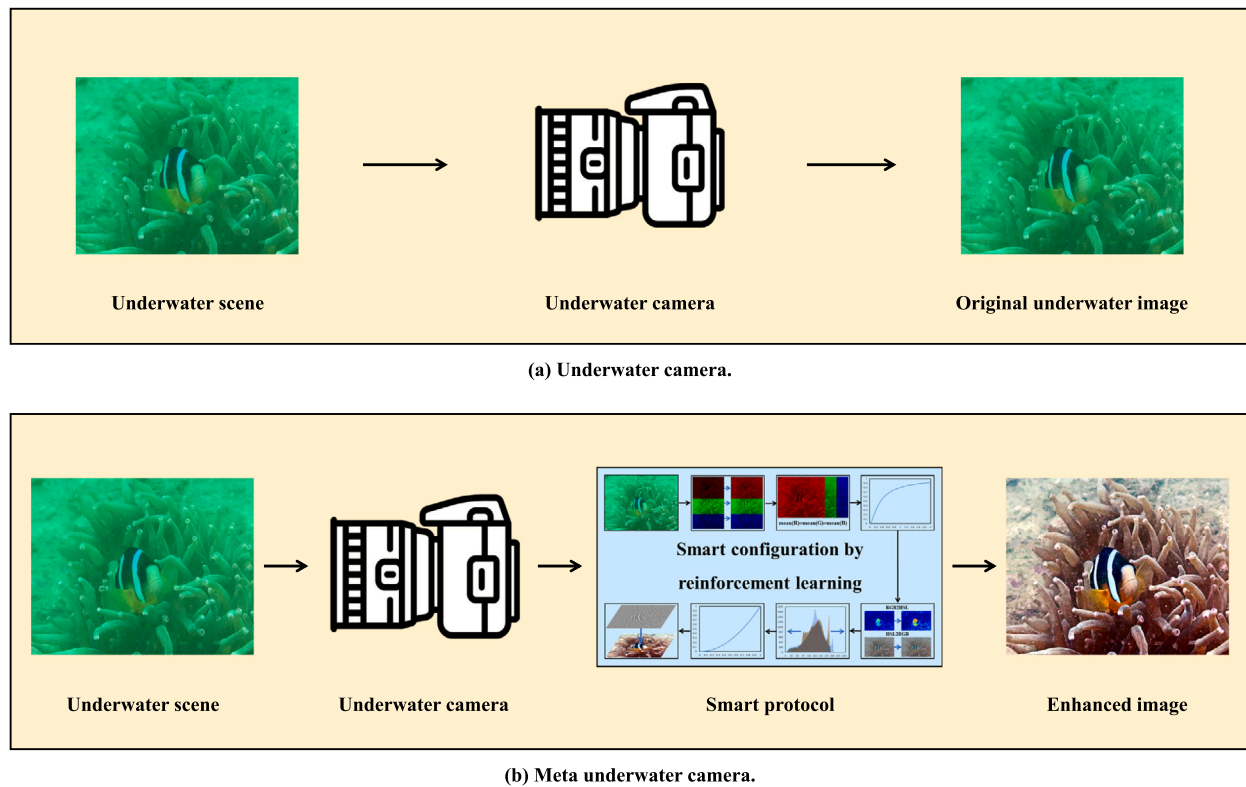


Fig. 4. Upgrade from an underwater camera to a meta underwater camera.

5.4. Methods for comparison

We conduct comprehensive evaluations among different methods. The proposed MUC methodology is compared with nine methods, i.e., UDCP (Drews et al., 2016), GBDRC (Li et al., 2016), IBLA (Peng and Cosman, 2017), fusion-based (Ancuti et al., 2012), CBAF (Ancuti et al., 2018), ACDC (Zhang et al., 2022b), UWCNN (Li et al., 2020a), Water-Net (Li et al., 2020b), and FUNIE-GAN (Islam et al., 2020a). Among them, UDCP (Drews et al., 2016), GBDRC (Li et al., 2016), and IBLA (Peng and Cosman, 2017) are physical model based methods, fusion-based (Ancuti et al., 2012), CBAF (Ancuti et al., 2018), and ACDC (Zhang et al., 2022b) are non-physical methods, and UWCNN (Li et al., 2020a), Water-Net (Li et al., 2020b), and FUNIE-GAN (Islam et al., 2020a) are data-driven methods.

5.5. Comprehensive evaluation on UIEB

5.5.1. Qualitative evaluation

We sort the underwater images into five categories, i.e., color deviation scenes, downing-looking scenes, forward-looking scenes, low-scatter scenes, and high-scatter scenes. We sample several underwater images from them. The image enhancement results of different methods are shown in Figs. 5–9.

As described in Section 2.1, the attenuation of light causes the color deviations of the images, resulting in the greenish, bluish, and yellowish phenomena. The image enhancement results of the color deviation scenes are shown in Fig. 5. The UDCP and GBDRC remove the effect of hazy veiling light but aggravate the effect of color deviations. The IBLA, UWCNN, and FUNIE-GAN have little effect on these underwater images. The fusion-based and Water-Net eliminate color deviations in Img1 and Img3, but not in Img2. The CBAF, ACDC, and MUC effectively eliminate color deviations. The results of CBCA and ACDC are dark and low color saturation, respectively.

The image enhancement results of the downward-looking and the forward-looking scenes are shown in Figs. 6 and 7, respectively. The

fusion-based, CBAF, ACDC, Water-Net, and MUC remove the effects of hazy veiling light and color deviations. The IBLA and FUNIE-GAN have good enhancement results in Img5, Img7, and Img9, but poor results in Img4, Img6, and Img8. The UWCNN reduces a certain color deviations but has a less positive effect on removing hazy veiling light. The UDCP and GBDRC significantly remove the effect of hazy veiling light but aggravate the effect of color deviations.

The scattering reduces the contrast and produces hazy veiling light in underwater images. The scattering effect depends on the distance between the camera and the scene (Chen et al., 2022). The image enhancement results of the low-scatter scenes are shown in Fig. 8. The fusion-based, Water-Net, and MUC have good performance in removing hazy veiling light and eliminating color deviations. The CBAF and ACDC remove hazy veiling light and eliminate color deviations but have low saturation. The UDCP, UWCNN, and FUNIE-GAN have good enhancement results in Img11, Img10, and Img12, respectively. The IBLA reduces a certain color deviations in Img10 and Img12 but has less positive effect in removing hazy veiling light. The results of GBDRC are reddish in color.

Compared with the low-scatter scenes, it is more difficult to enhance the high-scatter scenes. The image enhancement results of the high-scatter scenes are shown in Fig. 9. The UDCP, GBDRC, IBLA, UWCNN and FUNIE-GAN cannot remove hazy veiling light or eliminate color deviations, while the other methods have better performance.

In addition, we compare the detail enhancement results of underwater images, as shown in Fig. 10. The GBDRC, UWCNN, and FUNIE-GAN introduce extra color artifacts that cause the fishes and letters to become blurred. The UDCP, IBLA, fusion-based, CBAF, and Water-Net are not obvious for detail enhancement. The ACDC and MUC clearly highlight the texture of the fishes and letters, and the result of the MUC methodology appears better saturation and contrast.

The image enhancement results by our MUC methodology and the reference images from the UIEB dataset are shown in Fig. 11. The images enhanced by our MUC methodology are better at removing hazy

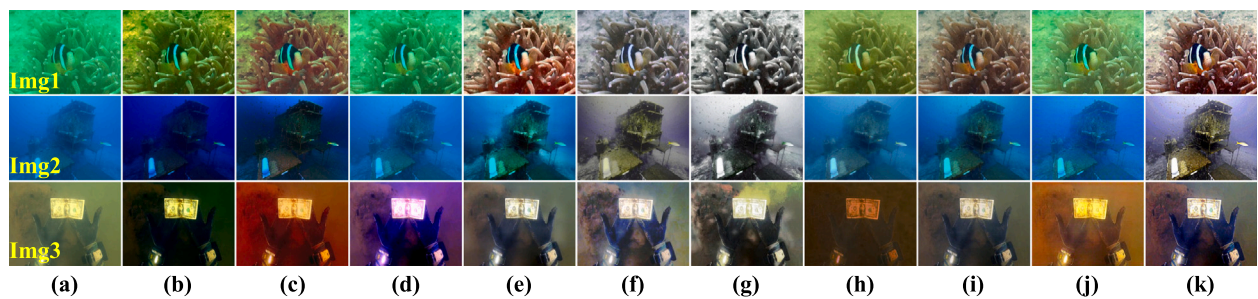


Fig. 5. The image enhancement results of the color deviation scenes from the UIEB dataset. (a) Underwater images, (b) UDCP (Drewns et al., 2016), (c) GBDRC (Li et al., 2016), (d) IBLA (Peng and Cosman, 2017), (e) Fusion-based (Ancuti et al., 2012), (f) CBAF (Ancuti et al., 2018), (g) ACDC (Zhang et al., 2022b), (h) UWCNN (Li et al., 2020a), (i) Water-Net (Li et al., 2020b), (j) FUNIE-GAN (Islam et al., 2020a), and (k) MUC (ours). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

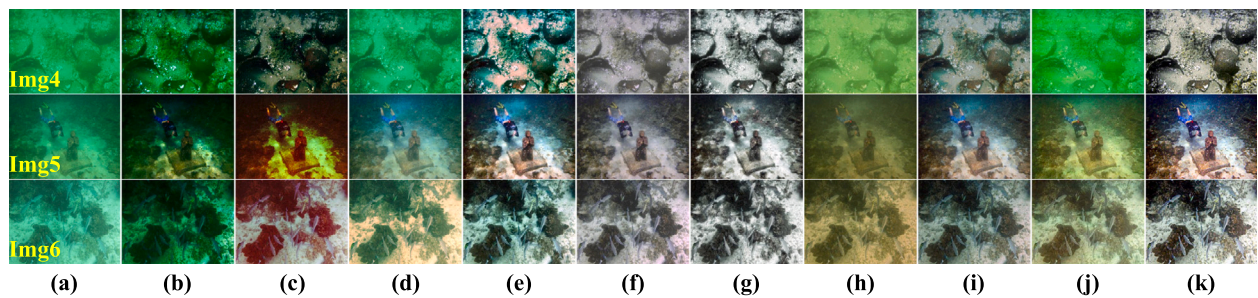


Fig. 6. The image enhancement results of the downward-looking scenes from the UIEB dataset. (a) Underwater images, (b) UDCP (Drewns et al., 2016), (c) GBDRC (Li et al., 2016), (d) IBLA (Peng and Cosman, 2017), (e) Fusion-based (Ancuti et al., 2012), (f) CBAF (Ancuti et al., 2018), (g) ACDC (Zhang et al., 2022b), (h) UWCNN (Li et al., 2020a), (i) Water-Net (Li et al., 2020b), (j) FUNIE-GAN (Islam et al., 2020a), and (k) MUC (ours).

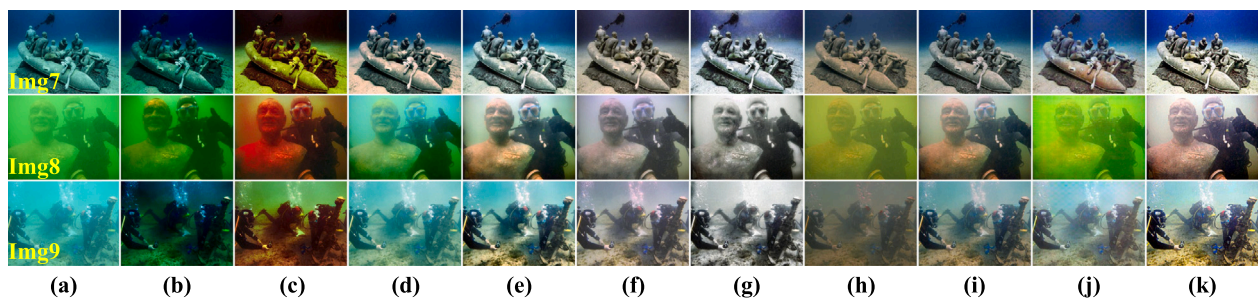


Fig. 7. The image enhancement results of the forward-looking scenes from the UIEB dataset. (a) Underwater images, (b) UDCP (Drewns et al., 2016), (c) GBDRC (Li et al., 2016), (d) IBLA (Peng and Cosman, 2017), (e) Fusion-based (Ancuti et al., 2012), (f) CBAF (Ancuti et al., 2018), (g) ACDC (Zhang et al., 2022b), (h) UWCNN (Li et al., 2020a), (i) Water-Net (Li et al., 2020b), (j) FUNIE-GAN (Islam et al., 2020a), and (k) MUC (ours).

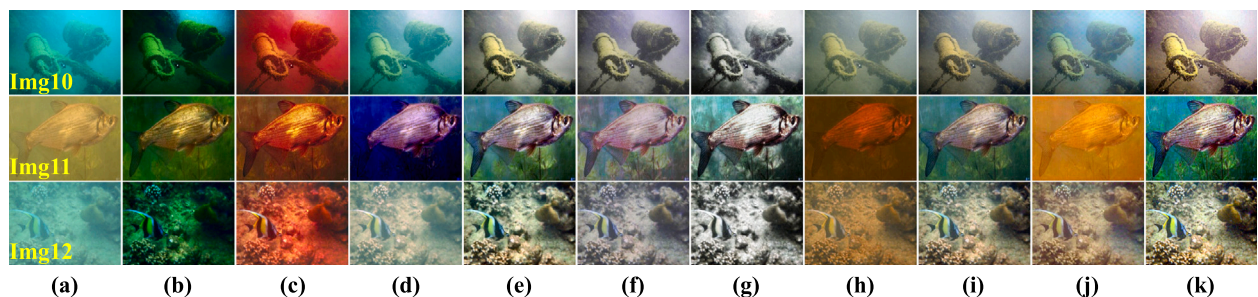


Fig. 8. The image enhancement results of the low-scatter scenes from the UIEB dataset. (a) Underwater images, (b) UDCP (Drewns et al., 2016), (c) GBDRC (Li et al., 2016), (d) IBLA (Peng and Cosman, 2017), (e) Fusion-based (Ancuti et al., 2012), (f) CBAF (Ancuti et al., 2018), (g) ACDC (Zhang et al., 2022b), (h) UWCNN (Li et al., 2020a), (i) Water-Net (Li et al., 2020b), (j) FUNIE-GAN (Islam et al., 2020a), and (k) MUC (ours).

veiling light (e.g., Img1, Img3, Img5, Img9, Img10, and Img11) and eliminating color deviations (e.g., Img2, Img4, Img6, Img7, and Img8).

The image enhancement results of the underwater images from the UIEB challenge set are shown in Fig. 12. The fusion-based method

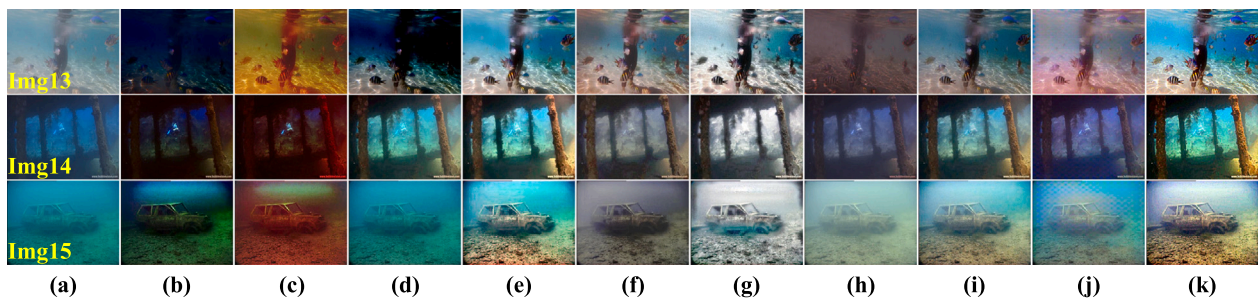


Fig. 9. The image enhancement results of the high-scatter scenes from the UIEB dataset. (a) Underwater images, (b) UDCP (Drews et al., 2016), (c) GBDRC (Li et al., 2016), (d) IBLA (Peng and Cosman, 2017), (e) Fusion-based (Ancuti et al., 2012), (f) CBAF (Ancuti et al., 2018), (g) ACDC (Zhang et al., 2022b), (h) UWCNN (Li et al., 2020a), (i) Water-Net (Li et al., 2020b), (j) FUNIE-GAN (Islam et al., 2020a), and (k) MUC (ours).

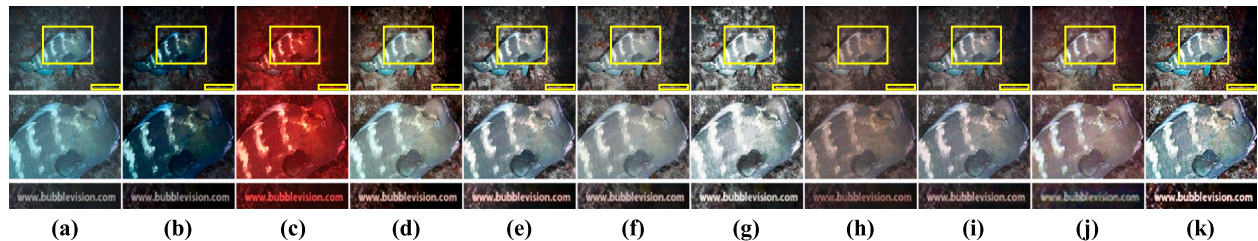


Fig. 10. The detailed image enhancement results of underwater images from UIEB dataset. (a) Underwater images, (b) UDCP (Drews et al., 2016), (c) GBDRC (Li et al., 2016), (d) IBLA (Peng and Cosman, 2017), (e) Fusion-based (Ancuti et al., 2012), (f) CBAF (Ancuti et al., 2018), (g) ACDC (Zhang et al., 2022b), (h) UWCNN (Li et al., 2020a), (i) Water-Net (Li et al., 2020b), (j) FUNIE-GAN (Islam et al., 2020a), and (k) MUC (ours).



Fig. 11. The image enhancement results by our MUC methodology and the reference images from the UIEB dataset. From top to bottom are underwater images, reference images from the UIEB dataset, and the image enhancement results of the MUC methodology (ours).

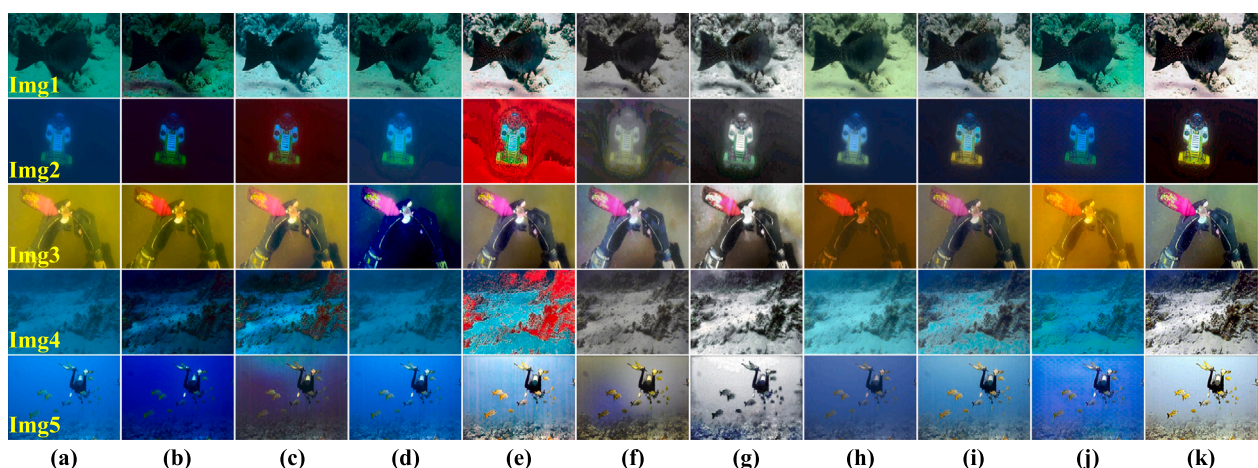


Fig. 12. The image enhancement results of the underwater images from the UIEB challenge set. (a) Underwater images, (b) UDCP (Drews et al., 2016), (c) GBDRC (Li et al., 2016), (d) IBLA (Peng and Cosman, 2017), (e) Fusion-based (Ancuti et al., 2012), (f) CBAF (Ancuti et al., 2018), (g) ACDC (Zhang et al., 2022b), (h) UWCNN (Li et al., 2020a), (i) Water-Net (Li et al., 2020b), (j) FUNIE-GAN (Islam et al., 2020a), and (k) MUC (ours).

introduces reddish artifacts (e.g., Img2 and Img4). The CBCA introduces white masks (e.g., Img2) and bluish artifacts (e.g., Img5). The other methods have limited capability of removing hazy veiling light

and eliminating color deviations. The ACDC and MUC produce visually pleasing results. It is worth noting that although our MUC methodology produces visually pleasing results for Img1 and Img2,

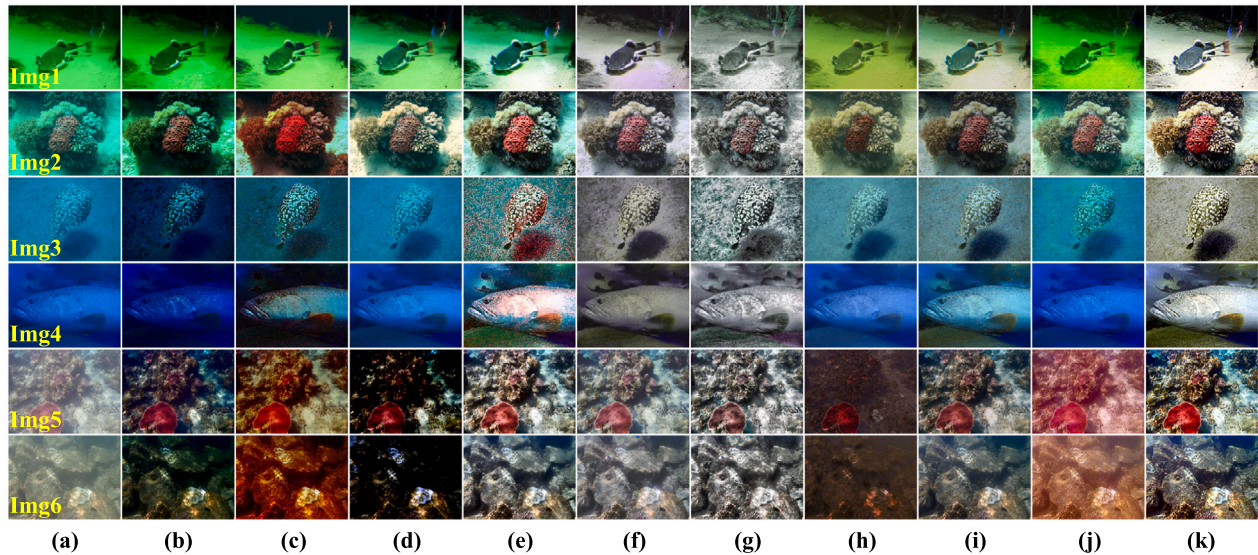


Fig. 13. The image enhancement results of the underwater images from the U45 dataset. (a) Underwater images, (b) UDCP (Drews et al., 2016), (c) GBDRC (Li et al., 2016), (d) IBLA (Peng and Cosman, 2017), (e) Fusion-based (Ancuti et al., 2012), (f) CBAF (Ancuti et al., 2018), (g) ACDC (Zhang et al., 2022b), (h) UWCNN (Li et al., 2020a), (i) Water-Net (Li et al., 2020b), (j) FUnIE-GAN (Islam et al., 2020a), and (k) MUC (ours).

Table 2

Quantitative evaluations of the image enhancement results of underwater images in Figs. 5–9 based on UCIQE and CCF metrics. The bold and blue figures indicate the first and second best results, respectively.

	UDCP		GBDRC		IBLA		Fusion-based		CBAF		ACDC		UWCNN		Water-Net		FUnIE-GAN		MUC (ours)	
	UCIQE↑	CCF↑	UCIQE↑	CCF↑	UCIQE↑	CCF↑	UCIQE↑	CCF↑	UCIQE↑	CCF↑	UCIQE↑	CCF↑	UCIQE↑	CCF↑	UCIQE↑	CCF↑	UCIQE↑	CCF↑	UCIQE↑	CCF↑
Img1	0.535	41.267	0.530	28.446	0.445	34.268	0.636	40.208	0.494	24.687	0.533	35.991	0.432	16.474	0.525	20.963	0.449	22.743	0.618	48.199
Img2	0.493	30.064	0.527	24.890	0.458	21.440	0.597	41.105	0.497	19.262	0.524	25.180	0.469	12.066	0.545	13.890	0.448	17.709	0.614	33.508
Img3	0.639	50.773	0.587	16.329	0.625	58.617	0.600	17.672	0.579	19.589	0.596	21.954	0.482	9.505	0.573	14.238	0.563	14.065	0.641	27.613
Img4	0.537	28.416	0.541	32.466	0.355	31.070	0.618	47.819	0.423	22.661	0.524	38.141	0.360	12.837	0.521	23.908	0.371	23.494	0.567	44.291
Img5	0.575	20.922	0.633	21.653	0.513	17.440	0.640	27.117	0.501	19.179	0.560	28.432	0.433	10.258	0.596	19.143	0.542	17.639	0.652	49.496
Img6	0.522	24.809	0.597	24.620	0.551	23.812	0.571	33.588	0.483	25.230	0.523	33.836	0.477	17.977	0.527	22.780	0.501	22.417	0.574	42.559
Img7	0.534	30.307	0.636	38.305	0.640	54.475	0.623	32.050	0.582	28.567	0.587	36.539	0.562	19.788	0.588	24.547	0.607	25.503	0.645	43.606
Img8	0.501	53.798	0.623	16.563	0.568	60.160	0.625	28.066	0.485	20.900	0.522	28.902	0.419	10.379	0.589	19.344	0.469	17.673	0.628	33.182
Img9	0.551	20.644	0.600	22.436	0.469	17.156	0.621	29.551	0.502	21.578	0.531	28.826	0.418	9.827	0.556	18.556	0.519	17.389	0.639	40.951
Img10	0.585	26.727	0.584	15.048	0.485	15.916	0.564	23.382	0.497	19.710	0.529	26.788	0.455	12.293	0.544	16.242	0.531	15.105	0.600	29.594
Img11	0.571	54.298	0.575	17.129	0.594	53.109	0.610	26.946	0.488	17.447	0.599	31.825	0.462	11.355	0.569	18.308	0.377	9.947	0.650	38.218
Img12	0.530	29.161	0.558	19.771	0.442	16.519	0.591	36.633	0.449	21.213	0.528	37.099	0.486	14.878	0.569	24.479	0.503	19.964	0.616	42.612
Img13	0.429	15.559	0.507	14.020	0.573	37.789	0.616	24.576	0.522	18.113	0.594	30.233	0.382	8.422	0.549	15.874	0.504	14.463	0.659	39.406
Img14	0.467	25.370	0.495	14.933	0.586	29.816	0.635	29.221	0.485	18.046	0.559	29.306	0.444	11.619	0.578	20.592	0.479	13.995	0.656	48.437
Img15	0.568	40.438	0.535	15.479	0.439	13.328	0.631	33.362	0.454	18.962	0.529	26.859	0.409	10.414	0.557	17.418	0.481	14.923	0.645	42.214

it introduces a few reddish artifacts. This implies one limitation of our MUC methodology in the low light situation (e.g., Img2), which also remains a challenge in existing studies for underwater image enhancement (Zhang et al., 2022c).

5.5.2. Quantitative evaluation

We evaluate the enhancement results of the underwater images in Figs. 5–9 by using UCIQE, CCF, UIQM, PCQI, BC(e), and BC(r), as shown in Tables 2–4. Our MUC methodology achieves the first best or the second best results on all the metrics for most images. The fusion-based method, IBLA, and UDCP achieve good results on UCIQE, CCF, and BC(e), respectively. The ACDC achieves good results on UIQM, PCQI, and BC(r). The above results are consistent with human visual perception.

In addition, the mean evaluation results of the UIEB dataset are shown in Table 5. Our MUC methodology has the first best results on UCIQE, CCF, PCQI, BC(e), and BC(r), and has the second best results on UIQM. The fusion-based, IBLA and UDCP have the second best results on UCIQE, CCF, and BC(e), respectively. The ACDC has the first best results on UIQM and the second best results on PCQI and BC(r).

Similarly, our MUC methodology achieves good performance over the mean evaluation results of the UIEB challenge set, as shown in

Table 6. In contrast to the other methods that are good on one metric but bad on the other metrics, the ACDC has good performance on four metrics, and our MUC methodology has consistent performance on all the metrics.

5.6. Comprehensive evaluation on U45

5.6.1. Qualitative evaluation

We sample several underwater images from three subsets of U45, as shown in Fig. 13. Among them, Img1 and Img2, Img3 and Img4, and Img5 and Img6 are sampled from the green subset, blue subset, and haze-like subset, respectively. The CBAF, ACDC, and MUC effectively eliminate color deviation both in greenish and bluish underwater images, and remove hazy veiling light. The fusion-based method removes hazy veiling light but does not completely eliminate green deviations (e.g., Img1 and Img2) and introduces reddish artifacts (e.g., Img3 and Img4). The Water-Net removes hazy veiling light but does not completely eliminate bluish deviations (e.g., Img3 and Img4). The UDCP significantly removes hazy veiling light but aggravates the color deviations. The GBDRC, IBLA, UWCNN, and FUnIE-GAN cannot eliminate color deviations and introduce reddish artifacts (e.g., Img5 and Img6).

Table 3

Quantitative evaluations of the image enhancement results of underwater images in Figs. 5–9 based on UIQM and PCQI metrics. The bold and blue figures indicate the first and second best results, respectively.

	UDCP		GBDRC		IBLA		Fusion-based		CBAF		ACDC		UWCNN		Water-Net		FUnIE-GAN		MUC (ours)	
	UIQM↑	PCQI↑	UIQM↑	PCQI↑	UIQM↑	PCQI↑	UIQM↑	PCQI↑	UIQM↑	PCQI↑	UIQM↑	PCQI↑	UIQM↑	PCQI↑	UIQM↑	PCQI↑	UIQM↑	PCQI↑	UIQM↑	PCQI↑
Img1	4.235	1.060	4.540	1.096	3.925	1.044	4.763	1.201	4.459	1.087	4.852	1.153	3.645	0.776	4.438	0.963	3.949	0.826	4.765	1.207
Img2	2.234	0.680	3.217	0.897	1.688	0.969	2.891	0.899	3.486	1.013	3.725	0.926	2.269	0.785	2.578	0.825	2.661	0.752	4.003	1.106
Img3	2.873	0.703	2.762	0.779	2.496	0.868	2.407	0.807	3.431	1.007	3.532	0.933	2.370	0.515	2.852	0.784	2.711	0.747	3.588	0.909
Img4	3.411	1.059	4.213	1.060	3.147	1.083	4.917	1.305	4.558	1.103	5.340	1.247	3.406	0.729	5.227	1.152	3.939	0.706	5.234	1.329
Img5	3.547	0.798	4.385	0.944	3.938	1.082	4.810	1.136	4.413	1.078	4.904	1.166	3.125	0.638	4.637	1.056	4.281	0.853	4.913	1.224
Img6	3.455	0.672	4.682	1.028	4.502	1.106	5.089	1.171	4.709	1.077	5.094	1.187	4.267	0.849	5.010	1.000	4.495	0.928	4.999	1.280
Img7	3.105	0.688	3.800	0.864	3.765	1.039	3.724	0.940	3.782	0.886	4.177	1.020	3.778	0.732	3.961	0.881	4.038	0.739	4.143	1.053
Img8	2.833	0.776	3.481	0.759	2.528	0.980	3.341	0.937	2.918	0.940	3.808	0.990	1.803	0.684	3.221	0.908	2.885	0.684	3.931	0.969
Img9	2.996	0.660	4.163	0.878	3.272	1.044	4.014	1.040	3.693	0.962	4.263	1.070	2.616	0.506	3.828	0.875	3.365	0.764	4.346	1.109
Img10	2.786	0.650	3.359	0.818	2.826	1.020	3.350	1.010	3.419	1.000	3.946	1.079	2.505	0.823	3.335	0.962	2.930	0.784	4.000	1.132
Img11	3.644	0.913	3.838	0.961	2.762	0.775	4.141	1.129	4.011	1.141	4.691	1.199	2.701	0.545	4.195	1.030	2.524	0.770	4.892	1.171
Img12	3.640	0.824	4.276	1.010	3.303	1.119	4.835	1.262	3.962	1.128	4.868	1.246	3.804	0.919	4.597	1.192	4.109	1.045	5.078	1.270
Img13	2.611	0.477	3.586	0.932	2.526	0.742	3.692	1.057	3.675	0.998	4.520	1.084	2.589	0.499	3.692	0.883	3.147	0.778	4.611	1.117
Img14	3.557	0.796	3.527	0.767	3.788	1.121	3.943	1.054	3.431	1.007	3.949	1.016	2.420	0.847	3.854	1.031	3.325	0.863	4.465	1.086
Img15	4.031	1.010	4.127	1.052	3.142	1.046	4.195	1.162	3.279	0.972	4.133	1.072	2.305	0.700	3.992	1.035	3.441	0.766	4.354	1.163

Table 4

Quantitative evaluations of the image enhancement results of underwater images in Figs. 5–9 based on BC(e) and BC(r) metrics. The bold and blue figures indicate the first and second best results, respectively.

	UDCP		GBDRC		IBLA		Fusion-based		CBAF		ACDC		UWCNN		Water-Net		FUnIE-GAN		MUC (ours)	
	BC(e)↑	BC(r)↑	BC(e)↑	BC(r)↑	BC(e)↑	BC(r)↑	BC(e)↑	BC(r)↑	BC(e)↑	BC(r)↑	BC(e)↑	BC(r)↑	BC(e)↑	BC(r)↑	BC(e)↑	BC(r)↑	BC(e)↑	BC(r)↑	BC(e)↑	BC(r)↑
Img1	1.162	1.601	1.118	1.641	0.374	1.252	1.279	2.692	0.930	2.002	1.265	2.797	0.062	1.218	1.010	1.717	0.131	1.634	1.438	3.414
Img2	2.296	1.085	3.374	2.259	0.689	1.334	0.931	2.213	1.380	3.280	1.645	3.928	0.152	1.697	0.609	1.603	0.114	1.520	1.966	5.486
Img3	2.028	1.181	0.532	1.078	1.466	1.682	0.471	1.615	1.569	2.689	2.130	2.870	0.251	0.831	0.658	1.496	0.050	1.506	2.253	2.899
Img4	0.865	2.031	1.372	2.215	0.447	1.389	1.367	4.534	0.857	2.305	1.307	4.187	0.241	1.306	1.348	2.895	0.075	1.578	1.414	5.407
Img5	1.391	1.114	1.473	1.390	0.869	1.548	1.323	2.589	1.042	2.473	1.309	3.271	0.184	1.074	1.322	2.136	0.749	2.122	1.581	3.900
Img6	0.753	0.947	0.627	1.437	0.492	1.512	0.792	2.145	0.655	1.656	0.727	2.325	0.351	1.151	0.853	1.689	0.426	1.632	0.856	3.054
Img7	0.403	0.834	0.531	1.080	0.168	1.314	0.043	1.558	0.033	1.271	0.426	2.097	0.063	0.997	0.113	1.293	0.101	1.146	0.427	2.382
Img8	1.557	1.517	4.582	1.481	2.417	1.677	4.499	2.431	3.048	2.403	6.771	3.428	0.218	1.336	4.204	2.165	0.132	1.609	7.745	4.130
Img9	1.540	1.297	1.316	1.489	0.476	1.379	1.017	2.575	0.726	1.932	1.140	3.036	0.024	0.815	0.962	1.816	0.090	1.624	1.295	3.698
Img10	1.993	1.169	1.182	1.205	0.549	1.276	1.222	2.346	1.183	2.043	2.123	3.143	0.091	1.364	1.230	1.879	0.233	1.559	2.345	3.690
Img11	6.809	2.039	6.805	2.159	9.045	1.937	6.585	3.716	5.603	3.130	9.323	5.171	2.108	0.923	7.108	2.943	0.267	1.234	11.020	6.210
Img12	14.704	1.833	8.978	1.933	2.208	1.592	11.784	3.717	6.626	2.287	12.205	4.053	4.910	1.755	10.418	2.832	6.845	2.363	14.969	4.896
Img13	4.108	0.896	2.131	1.603	4.117	2.236	2.405	2.855	2.084	2.367	3.846	4.651	1.249	1.047	2.530	2.032	0.442	1.581	4.542	4.739
Img14	2.685	1.295	2.415	1.217	2.293	2.676	1.937	3.532	1.690	3.117	2.375	3.957	0.386	1.302	2.183	2.881	0.946	1.867	3.682	4.997
Img15	1.526	2.181	1.203	2.216	0.272	1.354	0.823	4.685	0.479	1.998	0.808	4.692	0.262	1.260	0.709	2.850	0.003	1.614	1.008	5.814

Table 5

The mean evaluation results of the UIEB dataset based on UCIQE, CCF, UIQM, PCQI, BC(e), and BC(r) metrics. The bold and blue figures indicate the first and second best results, respectively.

Method	UCIQE ↑	CCF ↑	UIQM ↑	PCQI ↑	BC(e) ↑	BC(r) ↑
UDCP	0.589	36.926	3.498	0.814	1.467	1.287
GBDRC	0.604	26.252	3.975	0.860	1.237	1.325
IBLA	0.593	41.970	3.164	0.942	0.537	1.401
Fusion-based	0.635	34.767	4.193	1.023	0.943	2.632
CBAF	0.557	25.283	4.233	1.002	0.903	1.930
ACDC	0.569	32.455	4.559	1.062	1.341	2.943
UWCNN	0.480	13.147	3.508	0.587	0.219	0.948
Water-Net	0.578	21.301	4.244	0.897	0.899	1.759
FUnIE-GAN	0.562	21.917	3.979	0.758	0.120	1.392
MUC (ours)	0.652	44.736	4.272	1.113	1.681	3.229

Table 6

The mean evaluation results of the UIEB challenge set based on UCIQE, CCF, UIQM, PCQI, BC(e), and BC(r) metrics. The bold and blue figures indicate the first and second best results, respectively.

Method	UCIQE ↑	CCF ↑	UIQM ↑	PCQI ↑	BC(e) ↑	BC(r) ↑
UDCP	0.530	37.131	2.918	0.818	2.224	1.261
GBDRC	0.564	21.863	3.412	0.921	1.460	1.428
IBLA	0.571	35.816	2.983	0.994	0.661	1.740
Fusion-based	0.618	27.394	3.575	1.015	1.208	2.917
CBAF	0.515	19.729	3.667	1.027	1.968	2.674
ACDC	0.564	26.843	3.991	1.031	2.306	3.487
UWCNN	0.473	12.214	2.891	0.761	0.403	1.173
Water-Net	0.561	17.422	3.576	0.955	1.346	2.023
FUnIE-GAN	0.528	18.583	3.463	0.834	0.295	1.522
MUC (ours)	0.638	41.811	3.711	1.070	3.783	3.607

5.6.2. Quantitative evaluation

We calculate the mean evaluation results on the U45 and on its three subsets using UCIQE, CCF, UIQM, PCQI, BC(e), and BC(r), as shown in Table 7. Our MUC methodology has the first best results on CCF, BC(e), and BC(r), and has the second best results on UCIQE, UIQM, and PCQI.

5.7. Comprehensive evaluation on the standard color chart

To evaluate the performance of the different methods in eliminating color deviations, we qualitatively and quantitatively evaluate the image enhancement results with a standard color chart. The image enhancement results of the standard color chart are shown in Fig. 14. The results of CBAF and MUC are closer to the standard color chart. The

result of the fusion-based method suffers from over-enhancement and gives rise to overexposure. The UDCP, IBLA, UWCNN, Water-Net, and FUnIE-GAN have limited capability of eliminating color deviations.

To quantitatively evaluate the performance of the standard color chart enhancement results, we use the mean chroma difference ΔC^* and the mean color difference ΔE^* of standard color chart blocks (Lu et al., 2021) to evaluate the enhancement methods. The quantitative evaluation results are shown in Table 8. Our MUC methodology gives the first best results on ΔC^* and performs well in eliminating color deviations. In addition, the CBAF gives the first best results on ΔE^* and the MUC methodology gives the second best results on ΔE^* because the mean color difference ΔE^* calculates not only the chroma difference but also the brightness difference. The MUC methodology produces

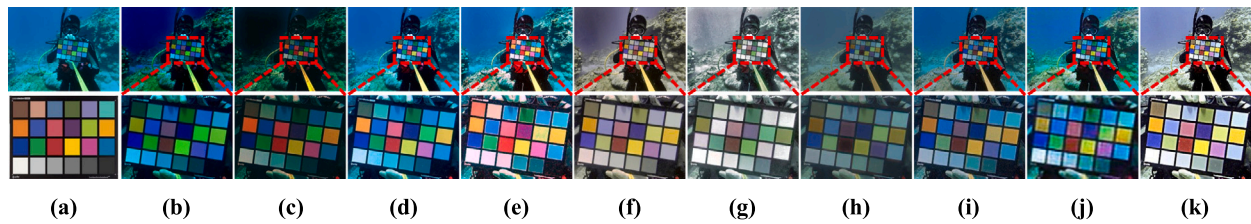


Fig. 14. The image enhancement results of the standard color chart. (a) Top row: underwater image. Down row: standard color chart, (b) UDCP (Drews et al., 2016), (c) GBDRC (Li et al., 2016), (d) IBLA (Peng and Cosman, 2017), (e) Fusion-based (Ancuti et al., 2012), (f) CBAF (Ancuti et al., 2018), (g) ACDC (Zhang et al., 2022b), (h) UWCNN (Li et al., 2020a), (i) Water-Net (Li et al., 2020b), (j) FUNIE-GAN (Islam et al., 2020a), and (k) MUC (ours).

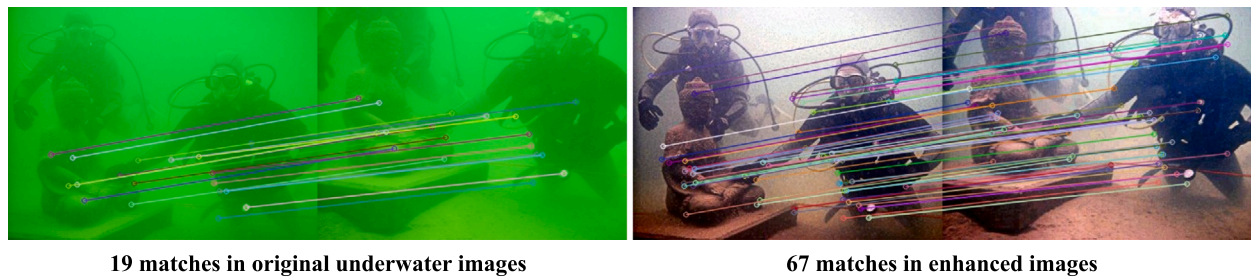


Fig. 15. SIFT local keypoint matching on original underwater images and enhanced images.

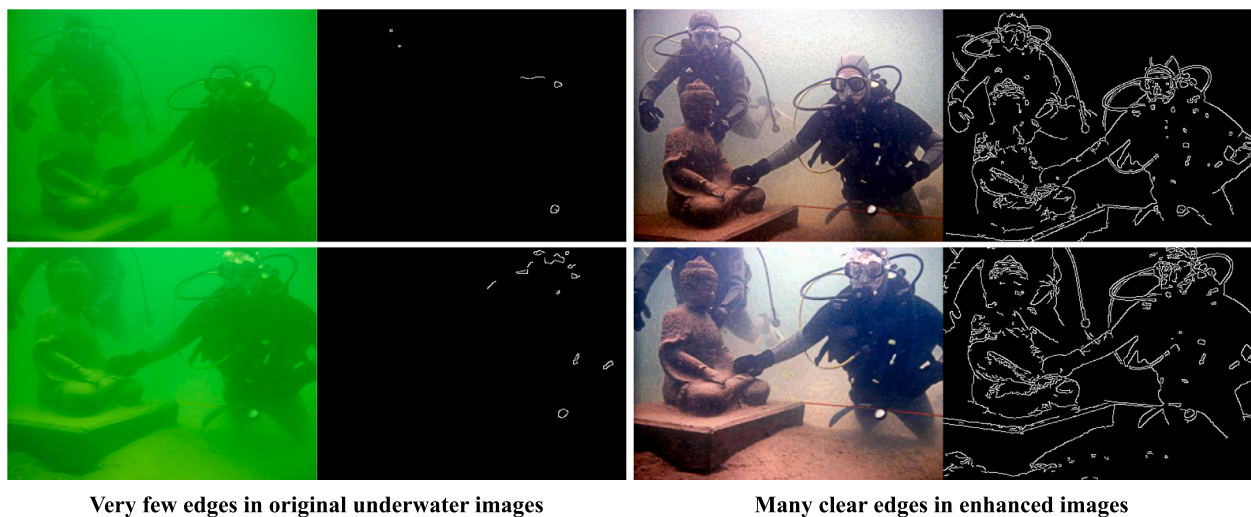


Fig. 16. Canny edge detection on original underwater images and enhanced images.

visually pleasing results and higher brightness than the results of CBAF as shown in Fig. 13. Therefore, the results of MUC have greater ΔE^* than the results of CBAF.

5.8. Runtime

Runtime is an indispensable quantitative evaluation metric for a method in practical application. We evaluate the runtime using the UIEB dataset and resize the 890 images to 256×256 , 512×512 , 768×768 , and 1024×1024 patches, separately. The mean runtime of executing 890 times of different methods on images of different resolutions is shown in Table 9. Our MUC methodology is significantly faster than other methods except FUNIE-GAN. Note that, limited by the network structure, FUNIE-GAN can only process images with a resolution of 256×256 . Although our MUC methodology is slower than FUNIE-GAN, it outperforms FUNIE-GAN in terms of the enhancement quality of underwater images and is not limited by image resolutions.

5.9. Applications

The degraded image quality can negatively affect the performance of various devices in underwater scenes (Yuan et al., 2022). High quality underwater images are highly demanded in many practical applications. We apply our MUC methodology to four tasks and explore its uses to these applications.

Firstly, we adopt scale-invariant feature transform (SIFT) local keypoint matching method (Lowe, 2004) to establish the correspondences between two similar scenes. As shown in Fig. 15, the number of matched keypoints of the enhanced images yields a nearly four-fold increase over those original matches. This validates that our MUC methodology can effectively enhance the local features of underwater images.

Secondly, we adopt Canny edge detection (Canny, 1986) to detect edge features. As shown in Fig. 16, the enhanced images render more edge features than the original underwater images. This validates that our MUC methodology can effectively recover local features from underwater images.

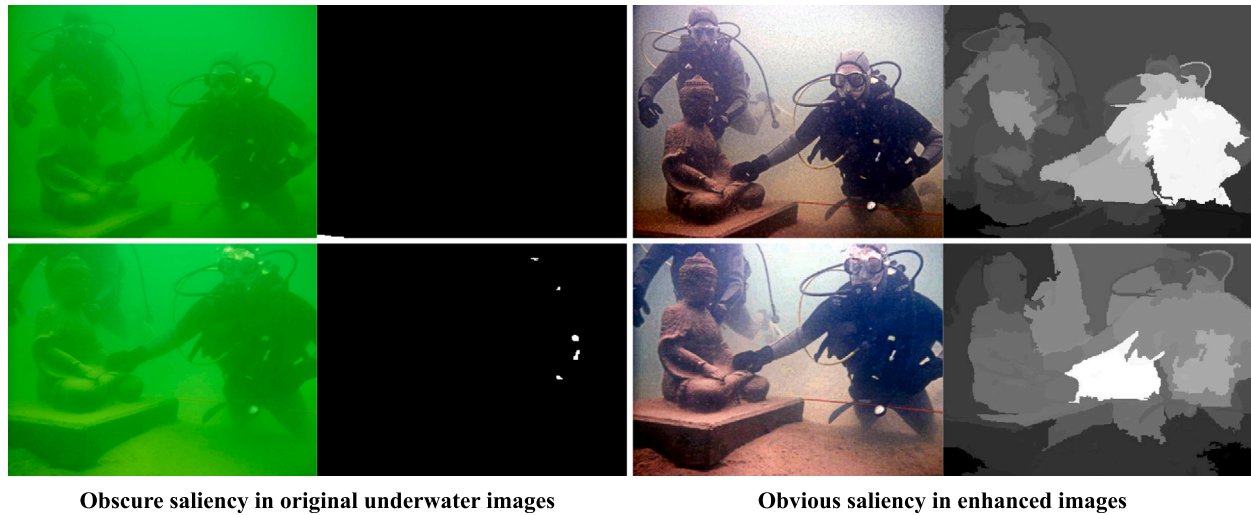


Fig. 17. Saliency detection on original underwater images and enhanced images.

Table 7

The mean evaluation results on the U45 dataset and on its three subsets based on UCIQE, CCF, UIQM, PCQI, BC(e), and BC(r) metrics. The bold and blue figures indicate the first and second best results, respectively.

Subset	Method	UCIQE ↑	CCF ↑	UIQM ↑	PCQI ↑	BC(e) ↑	BC(r) ↑
Green	UDCP	0.523	36.980	3.253	0.854	1.568	1.241
	GBDRC	0.583	22.055	3.856	0.852	2.950	1.357
	IBLA	0.546	45.145	3.147	1.036	0.773	1.373
	Fusion-based	0.622	35.843	4.008	1.078	3.213	3.338
	CBAF	0.516	23.205	3.823	1.071	1.853	2.004
	ACDC	0.520	28.550	4.661	1.229	4.677	3.628
	UWCNN	0.459	14.476	2.958	0.794	0.060	1.601
	Water-Net	0.573	21.084	3.989	1.010	2.721	2.202
	FUnIE-GAN	0.509	23.500	3.658	0.864	0.703	1.636
	MUC (ours)	0.609	40.023	4.200	1.108	5.336	3.631
Blue	UDCP	0.589	34.662	2.939	0.793	1.381	1.209
	GBDRC	0.578	23.865	3.740	0.866	1.472	1.506
	IBLA	0.555	35.292	2.814	1.004	0.266	1.344
	Fusion-based	0.655	37.899	4.202	1.023	1.903	4.466
	CBAF	0.530	24.008	3.746	1.048	0.644	1.949
	ACDC	0.534	27.383	4.648	1.092	1.597	3.837
	UWCNN	0.484	15.151	3.017	0.750	0.105	1.209
	Water-Net	0.566	23.829	4.117	0.944	1.659	2.404
	FUnIE-GAN	0.544	23.865	3.612	0.900	0.308	1.540
	MUC (ours)	0.620	40.248	3.971	1.095	1.126	3.849
Haze-like	UDCP	0.599	40.086	4.098	0.852	2.470	1.631
	GBDRC	0.595	23.133	4.279	0.867	2.184	1.526
	IBLA	0.613	45.244	2.107	0.697	1.756	1.633
	Fusion-based	0.631	31.849	4.754	1.176	1.786	2.773
	CBAF	0.530	22.006	4.530	1.132	1.593	2.205
	ACDC	0.560	30.565	5.173	1.271	2.240	3.693
	UWCNN	0.444	10.576	3.714	0.508	1.385	0.893
	Water-Net	0.585	20.699	4.613	1.043	1.746	1.973
	FUnIE-GAN	0.536	17.675	4.091	0.907	0.959	1.609
	MUC (ours)	0.668	46.062	4.890	1.252	2.519	3.932
U45	UDCP	0.570	37.242	3.430	0.833	1.807	1.360
	GBDRC	0.585	23.018	3.958	0.862	2.202	1.463
	IBLA	0.571	41.894	2.689	0.912	0.931	1.450
	Fusion-based	0.636	35.197	4.321	1.093	2.301	3.526
	CBAF	0.525	23.073	4.033	1.083	1.363	2.053
	ACDC	0.538	28.833	4.827	1.197	2.838	3.719
	UWCNN	0.462	13.401	3.230	0.684	0.477	1.234
	Water-Net	0.574	21.871	4.240	0.999	2.042	2.193
	FUnIE-GAN	0.530	21.680	3.787	0.890	0.657	1.595
	MUC (ours)	0.632	42.111	4.353	1.151	2.994	3.804

Thirdly, we adopt saliency detection (Yan et al., 2013) to produce the saliency maps. As shown in Fig. 17, the enhanced images exhibit more saliency information than the original underwater images.

This validates that our MUC methodology can effectively enhance the saliency regions from underwater images.

Fourth, we conduct the underwater detection on the URPC2021¹ dataset with YOLOv5². As shown in Fig. 18, the MUC not only effectively eliminates color deviations, improves contrasts, and removes the detail blurs, but also achieves the best detection performance among different underwater image enhancement methods. Table 10 reports the detection accuracy on the enhanced results from the different underwater image enhancement methods. It can be seen that the MUC outperforms the comparison methods in terms of detection accuracy.

The four tasks illustrate the potential of our MUC methodology in broad applications.

5.10. Ablation study

5.10.1. Empirical validation of the order of the seven techniques

The comprehensive protocol comprises seven image enhancement techniques which sequentially solve the three problems of color deviations (CD), low contrasts (LC), and detail blurs (DB) arising from underwater images. To evaluate the performance of different orders of the seven techniques, we change the order of them for solving the three problems of CD, LC, and DB. Specifically, permutations of the seven techniques organized in orders of LC-CD-DB, LC-DB-CD, DB-CD-LC, DB-LC-CD, CD-DB-LC, and CD-LC-DB (ours) are tested, and the visualized results are given in Fig. 19. It can be seen that the results from alternative orders suffer from color deviations (i.e., DB-CD-LC and CD-DB-LC) and low contrasts (i.e., LC-CD-DB, LC-DB-CD, and DB-LC-CD). In contrast, our comprehensive protocol achieves the best performance.

The quantitative evaluations for different orders of the seven techniques are given in Table 11. Our comprehensive protocol achieves the first best results on UCIQE, CCF, UIQM, BC(e), and BC(r), and achieves the second best results on PCQI. The results validate the effectiveness of the order of the seven techniques adopted by our protocol.

5.10.2. Empirical validation of the effectiveness of the seven techniques

We validate the effectiveness of each technique in our MUC methodology by ablation study. This is done by gradually adding the image enhancement techniques to the protocol. Fig. 20 illustrates the intermediate results at each stage of our MUC methodology. The column w/ (a) illustrates the intermediate results by using the technique (a) presented in Section 2.1. We can see that at this stage certain green deviations are

¹ <http://urpc.pcl.ac.cn/>

² <https://github.com/ultralytics/yolov5>

Table 8

Quantitative evaluations of the image enhancement results of the standard color chart. The bold and blue figures indicate the first and second best results, respectively.

Index	UDCP	GBDRC	IBLA	Fusion-based	CBAF	ACDC	UWCNN	Water-Net	FUnIE-GAN	MUC (ours)
$\Delta C \downarrow$	39.318	16.325	27.355	20.416	16.671	28.500	24.472	23.900	27.432	15.413
$\Delta E \downarrow$	41.056	25.907	29.180	25.717	20.234	37.064	28.943	26.040	28.663	25.602

Table 9

The mean runtime of each method for different image resolutions (in seconds). The bold and blue figures indicate the first and second best results, respectively.

Resolution	UDCP	GBDRC	IBLA	Fusion-based	CBAF	ACDC	UWCNN	Water-Net	FUnIE-GAN	MUC (ours)
256 × 256	2.670	2.996	6.093	0.137	0.355	0.311	4.581	0.671	0.021	0.087
512 × 512	10.987	12.236	25.573	0.356	0.826	1.204	4.788	0.909	–	0.273
768 × 768	25.198	27.769	58.308	0.657	1.705	2.518	4.986	1.269	–	0.526
1024 × 1024	45.643	51.767	102.791	1.237	2.953	4.506	5.222	1.761	–	1.035

“–” indicates the result is not available.

Table 10

Detection accuracy on the enhanced results from different underwater image enhancement methods. The bold and blue figures indicate the first and second best results, respectively.

Index	UDCP	GBDRC	IBLA	Fusion-based	CBAF	ACDC	UWCNN	Water-Net	FUnIE-GAN	MUC (ours)
mAP (%)	66.3	63.0	64.2	64.3	64.3	62.8	65.0	64.9	63.8	68.6

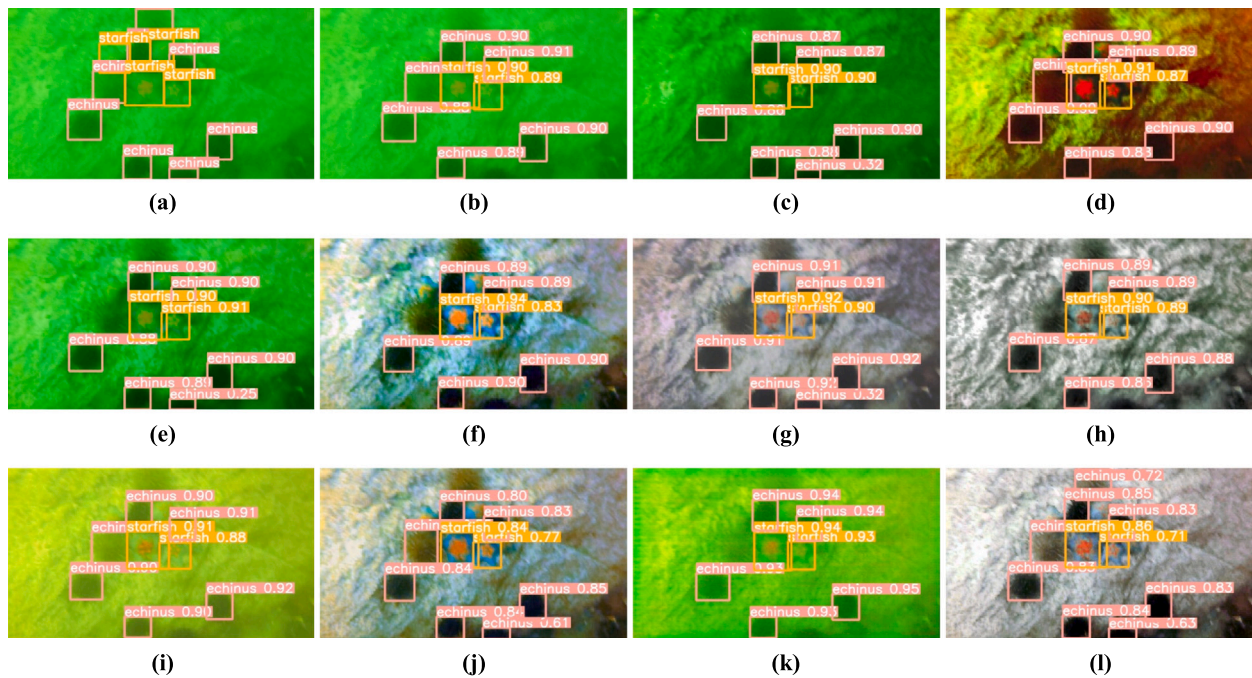


Fig. 18. The detection results for underwater images with different underwater image enhancement methods. (a) Ground truth, (b) underwater image, (c) UDCP (Drews et al., 2016), (d) GBDRC (Li et al., 2016), (e) IBLA (Peng and Cosman, 2017), (f) Fusion-based (Ancuti et al., 2012), (g) CBAF (Ancuti et al., 2018), (h) ACDC (Zhang et al., 2022b), (i) UWCNN (Li et al., 2020a), (j) Water-Net (Li et al., 2020b), (k) FUnIE-GAN (Islam et al., 2020a), and (l) MUC (ours).

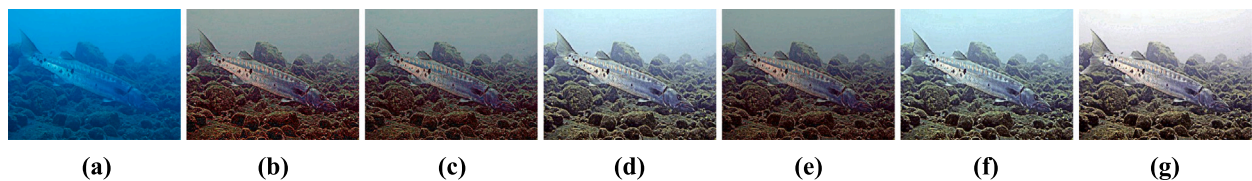


Fig. 19. Visualized underwater image enhancement results from the seven techniques in different orders. (a) Original image, (b) LC-CD-DB, (c) LC-DB-CD, (d) DB-CD-LC, (e) DB-CD-LC, (f) CD-DB-LC, (g) CD-LC-DB (ours).

eliminated. The column w/ (a)–(b) illustrates the intermediate results by using the techniques (a) and (b). We can see that at this stage green deviations are further eliminated. However, the intensity values for the

images may be greater than 1. To be able to display the image, we set intensity values greater than 1 to 1. The column w/ (a)–(c) illustrates the intermediate results by using the techniques (a), (b), and (c). We

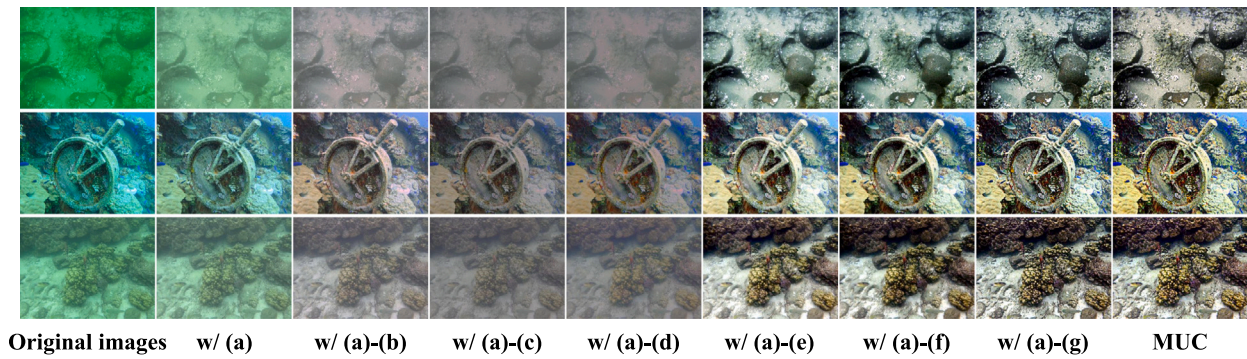


Fig. 20. Ablated results of our MUC methodology.

Table 11

The mean evaluation results of different orders of the seven techniques based on UCIQE, CCF, UIQM, PCQI, BC(e), and BC(r) metrics. The bold and blue figures indicate the first and second best results, respectively.

Order	UCIQE ↑	CCF ↑	UIQM ↑	PCQI ↑	BC(e) ↑	BC(r) ↑
LC-CD-DB	0.594	41.176	3.821	1.006	1.667	2.487
LC-DB-CD	0.577	37.663	3.789	0.987	1.627	2.417
DB-CD-LC	0.642	42.258	4.171	1.155	1.553	3.114
DB-LC-CD	0.573	36.968	3.714	0.978	1.551	2.310
CD-DB-LC	0.639	41.942	4.193	1.142	1.667	3.007
CD-LC-DB (ours)	0.644	44.441	4.232	1.142	1.682	3.229

can see that at this stage the brightness is reduced. The column w/ (a)–(d) illustrates the intermediate results by using the techniques (a), (b), (c), and (d). We can see that at this stage the color saturation is improved. The column w/ (a)–(e) illustrates the intermediate results by using the techniques (a), (b), (c), (d), and (e). We can see that at this stage the contrasts are improved and the haze veiling light is removed. The column w/ (a)–(f) illustrates the intermediate results by using the techniques (a), (b), (c), (d), (e), and (f). We can see that at this stage the contrasts of the high grayscale area are increased and the contrasts of the low grayscale area are reduced. The column w/ (a)–(g) illustrates the intermediate results by using the techniques (a), (b), (c), (d), (e), (f), and (g). We can see that at this stage the sharpness is improved and the details are enriched. The column MUC illustrates the intermediate results by using the overall MUC. We can see that at the final stage the color deviations are further corrected and the images are more in line with visual preference.

The quantitative evaluations for the ablation study of our MUC methodology are given in Table 12. Our MUC methodology achieves the first best results on UCIQE, CCF, UIQM, and BC(r), and achieves the second best results on PCQI and BC(e). The ablation study validates the necessity of every image enhancement techniques for enhancing visually preferable underwater images.

The ablation study of gradually adding the individual techniques to the protocol validates that the seven techniques address different issues regarding underwater degrade, and they play roles complementary to each other in enhancing underwater images. The seven techniques play an overall role in comprehensively enhancing underwater images.

5.10.3. Analysis of different feature extraction networks

We analyze the effects of using different feature extraction networks in our MUC methodology. Specifically, we test three networks, i.e., Resnet18, Resnet50, and VGG16, as the feature extraction networks. As shown in Table 13, the Resnet18, Resnet50, and VGG16 achieve similar results on the underwater image quality metrics (i.e., UCIQE, CCF, UIQM, PCQI, BC(e), and BC(r)). On the other hand, the Resnet18 is significantly faster than the other networks. Therefore, we use the Resnet18 as the feature extraction network in our MUC methodology.

Table 12

The mean evaluation results of the ablation study of our MUC methodology based on UCIQE, CCF, UIQM, PCQI, BC(e), and BC(r) metrics. The bold and blue figures indicate the first and second best results, respectively.

Techniques	UCIQE ↑	CCF ↑	UIQM ↑	PCQI ↑	BC(e) ↑	BC(r) ↑
w/ (a)	0.514	18.364	3.469	0.819	−0.107	1.087
w/ (a)–(b)	0.525	18.398	3.505	0.825	−0.086	1.111
w/ (a)–(c)	0.476	16.549	3.142	0.684	−0.200	0.907
w/ (a)–(d)	0.480	15.793	3.168	0.683	−0.196	0.914
w/ (a)–(e)	0.632	34.125	3.998	0.951	0.829	2.082
w/ (a)–(f)	0.645	38.571	4.000	0.986	1.073	2.007
w/ (a)–(g)	0.644	44.441	4.232	1.142	1.682	3.229
MUC	0.652	44.736	4.272	1.113	1.681	3.229

Table 13

The mean evaluation results of the different feature extraction networks based on UCIQE, CCF, UIQM, PCQI, BC(e), BC(r), and runtime (in seconds). The bold and blue figures indicate the first and second best results, respectively.

Networks	UCIQE ↑	CCF ↑	UIQM ↑	PCQI ↑	BC(e) ↑	BC(r) ↑	Runtime ↓
Resnet18	0.652	44.736	4.272	1.113	1.681	3.229	0.087
Resnet50	0.657	45.144	4.263	1.120	1.649	3.194	0.107
VGG16	0.646	44.433	4.245	1.138	1.684	3.229	0.113

Table 14

The mean evaluation results of the ablation study of the multi-color space features based on UCIQE, CCF, UIQM, PCQI, BC(e), and BC(r) metrics. The bold and blue figures indicate the first and second best results, respectively.

Techniques	UCIQE ↑	CCF ↑	UIQM ↑	PCQI ↑	BC(e) ↑	BC(r) ↑
w/o HSL	0.646	44.445	4.233	1.143	1.686	3.228
w/o LAB	0.643	44.190	4.231	1.138	1.670	3.229
w/o HSL+LAB	0.641	44.182	4.227	1.141	1.667	3.229
MUC	0.652	44.736	4.272	1.113	1.681	3.229

5.10.4. Analysis of multi-color space features

We test the effects of the multi-color space features of our MUC methodology. Specially, the ablated models includes: (a) multi-color space features without HSL (w/o HSL), (b) multi-color space features without LAB (w/o LAB), and (c) multi-color space features without HSL and LAB (w/o HSL+LAB). As shown in Table 14, our MUC methodology (i.e., multi-color space features) achieves the first best results on UCIQE, CCF, UIQM, and BC(r), and achieves the second best result on BC(e). In addition, removing any one of the three color space features decreases the performance.

5.10.5. Reinforcement learning reward analysis

We study the effect of the reinforcement learning reward on our MUC methodology. We set the reward to UCIQE, UIQM, and the combination of UCIQE and UIQM, separately. As shown in Fig. 21, the results of UCIQE have color deviations and over-saturation. The results of UIQM achieve better results than the results of UCIQE, but still have color deviations and low saturations. The results of the combination of

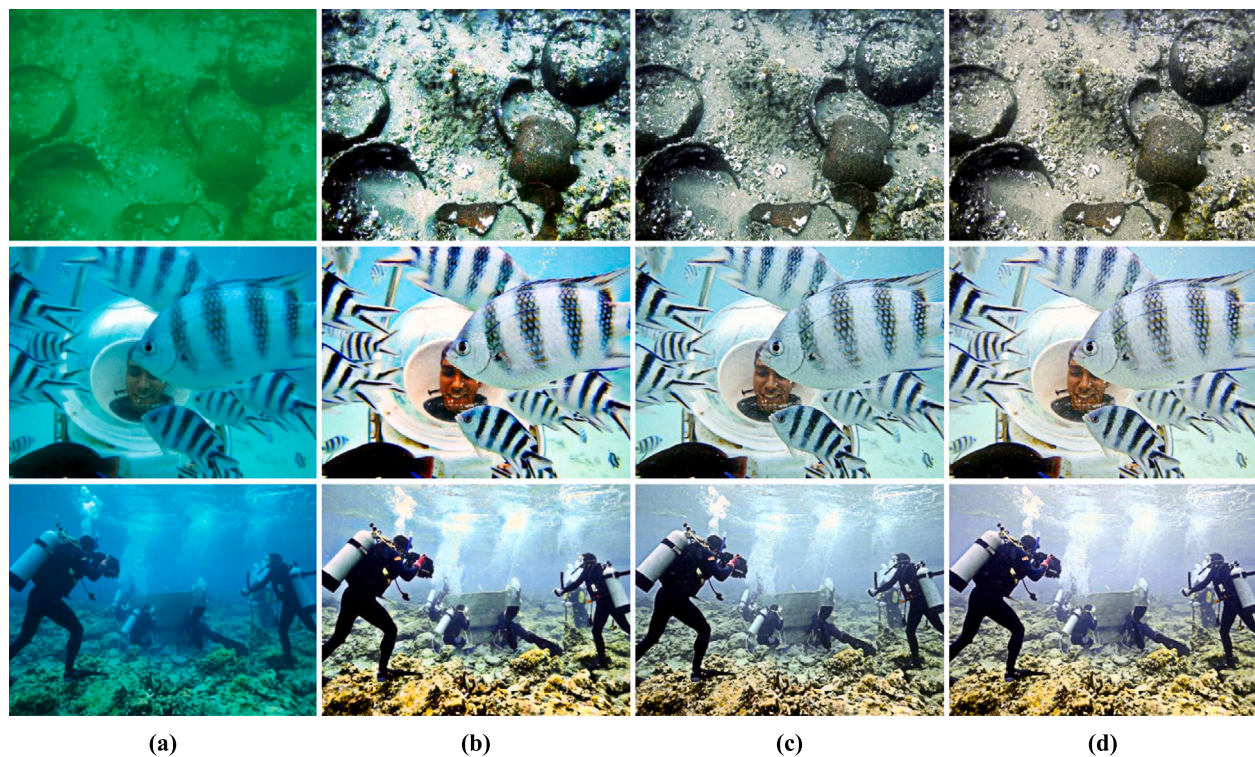


Fig. 21. The results based on different rewards of our MUC methodology. (a) Original images, (b) UCIQE, (c) UIQM, (d) the combination of UCIQE and UIQM (ours).

Table 15

The mean evaluation results of different settings of the balance parameter σ based on UCIQE, CCF, UIQM, PCQI, BC(e), and BC(r) metrics. The bold and blue figures indicate the first and second best results, respectively.

σ	UCIQE $\uparrow$	CCF $\uparrow$	UIQM $\uparrow$	PCQI $\uparrow$	BC(e) $\uparrow$	BC(r) $\uparrow$	Average $\uparrow$
0.35	0.651	44.766	4.242	1.114	1.628	3.189	9.265
0.45	0.646	44.442	4.235	1.141	1.681	3.229	9.229
0.55	0.649	44.748	4.241	1.113	1.621	3.190	9.260
0.65	0.652	44.736	4.272	1.113	1.681	3.229	9.281
0.75	0.644	44.438	4.231	1.142	1.681	3.229	9.228
0.85	0.643	44.191	4.229	1.139	1.666	3.227	9.183
0.95	0.651	44.763	4.244	1.113	1.632	3.189	9.265
1.05	0.615	37.119	4.235	1.049	1.187	3.365	7.928

UCIQE and UIQM achieve the color deviation elimination and suitable saturation. They are more in line with visual preference.

In addition, we study the effect of the balance parameter σ in Eq. (10) on our MUC methodology. We set it to 0.35, 0.45, 0.55, 0.65, 0.75, 0.85, 0.95, and 1.05, separately. The experiments are conducted on the UIEB (Li et al., 2020b) dataset using UCIQE, CCF, UIQM, PCQI, BC(e), and BC(r) as the evaluation metrics. As shown in Table 15, the balance parameter σ has significant effects on the performance of the MUC. When it set to 0.65, the MUC achieves stable and good performance. Therefore, we set σ to 0.65 in our experiments.

6. Conclusion

This paper has introduced a meta underwater camera (MUC) methodology. It operates behind an underwater camera's operation of capturing underwater images, providing underwater images of high visibility beyond those of low visibility originally obtained from the underwater camera. Our MUC methodology comprises a comprehensive cascade of seven image enhancement techniques for solving the problems of color deviations, low contrasts, and detail blurs with underwater images. In addition, it is smartly configured by reinforcement learning that provides optimal parameter values for the techniques to

achieve the optimal enhancement performance. Last but not least, it gives visually preferable enhanced images in favor of human perception. Comprehensive qualitative and quantitative empirical evaluations have validated that our MUC methodology outperforms nine underwater image enhancement methods, and our MUC methodology has extremely fast speed, which can achieve 0.087 s for a single 256×256 patch image. All these validate that our MUC methodology is practical for upgrading an underwater camera into a meta underwater camera which can provide underwater images of high visibility in an efficient way.

Although our methodology has good performance in enhancing the color, contrasts, and details of underwater images, it still cannot handle the underwater images acquired in low light conditions well. In addition, the existing image quality evaluation metrics designed for underwater image have limitations in some cases and limit the performance of our methodology. Therefore, our proposed methodology is a generic methodology, and the implementation scheme in our work is just one option for implementing the methodology. More effective image processing algorithms and image quality evaluation metrics can be employed or developed for enriching the proposed methodology.

Additionally, as the underwater image enhancement techniques characterized by our MUC methodology can change images into unseen representations, using our MUC methodology to enrich the training data may improve the robustness of the tasks such as underwater image classification and underwater object detection. In the future, we will explore the possibility of using our MUC methodology as a data augmentation method for the purpose of increasing the representational power of learning models.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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